

Minkowski Bounds for Pasten’s Arithmetic Derivative Lattices in the Squarefree Subfamily

Anonymous (verification kernel project)

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Abstract

For each coprime triple (a, b, c) with $a + b = c$, Pasten (2021) constructs a *universal derivative lattice* $F(a, b)$ of rank $\omega(abc) - 1$ from the linear constraint imposed by the weighted arithmetic derivative. We prove that for *squarefree* coprime triples, the lattice determinant satisfies $\det(F(a, b)) < \text{rad}(abc)$. Combined with Vaaler’s subspace lattice theorem, this yields a nonzero lattice vector with ℓ^∞ -norm $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$.

We give a structural classification of partition types by whether the minimum *non-degenerate* norm ratio $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$ is universally bounded: nine types are bounded. We prove unconditional upper bounds, including $2^{-1/(\omega-1)}$ for balanced types at $\omega = 3, 4, 5$, $\rho < 1$ for two $\omega = 5$ types with $a = 1$, $\rho < 2^{-1/4}$ for type $(2, 1, 2)$, and $\rho < 6^{-1/4}$ for type $(2, 2, 1)$. For $\omega \geq 6$ we prove an explicit unboundedness criterion; applying it to every partition type requires the corresponding infinite prime-pattern family. The asymptotic constants are sharp only under the relevant prime-pattern conjectures. The inequalities have been checked over 44 000 squarefree triples and the key inequalities formalized in Lean 4 (zero sorry).

We prove a clean quality boundary: for squarefree triples with $a \leq b$, quality $> 1/2$ if and only if $a = 1$. Combined with the classification, this shows that the triples hardest for abc (highest quality) are structurally opposite to the triples closest to the extremal ρ bound in the surveyed families. The argument is self-contained and uses no abc-equivalent hypothesis.

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Honest scope. The result proved here is a structural property of Pasten’s lattice. It does *not* prove the abc conjecture, nor does it provide progress toward abc for the hard cases (high-quality triples, which have small ω). The abc conjecture remains [OUT] in the sense of the accompanying verification kernel.

1 Introduction

1.1 Background

The abc conjecture of Masser–Oesterlé asserts that for every $\varepsilon > 0$ there exists $K_\varepsilon > 0$ such that for all coprime positive integers a, b, c with $a + b = c$:

$$c \leq K_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

where $\text{rad}(n) = \prod_{p|n} p$ is the radical. This remains an open problem.

Pasten [1] introduced a framework relating the abc conjecture to the geometry of numbers via the *universal arithmetic derivative*. For each prime p and weight $\psi(p) \in \mathbb{Z}$, the ψ -arithmetic derivative is

$$d^\psi(n) = n \sum_{p|n} \frac{v_p(n)}{p} \psi(p),$$

where $v_p(n)$ is the p -adic valuation. The key object is the *lattice of additive weights*:

$$F(a, b) = \{ \psi : \mathbb{P}(abc) \rightarrow \mathbb{Z} \mid d^\psi(a) + d^\psi(b) = d^\psi(c) \}.$$

Since $\gcd(a, b) = 1$ and $a + b = c$, the prime supports P_a, P_b, P_c of a, b, c are pairwise disjoint (for squarefree triples), and the additivity constraint reduces to a single integer linear equation. Pasten showed that $F(a, b)$ has rank $\omega(abc) - 1$, where $\omega(n)$ denotes the number of distinct prime divisors of n .

Pasten's *Small Derivatives Conjecture* (SDC) asserts that, apart from finitely many triples and from triples of the form $(1, N, q)$ with q prime (up to order), the minimum ℓ^∞ -norm of a non-degenerate vector in $F(a, b)$ is at most c^η for some $\eta < 1$. Pasten proved that SDC and the abc conjecture are equivalent with an explicit relation of exponents. Siegel's lemma gives only $\eta = 1$ (the trivial bound); beating this would be a breakthrough.

Our contribution is more modest: we establish the exact determinant of $F(a, b)$ for squarefree triples, and derive the corresponding Minkowski bound $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$. This is the $\eta = 1/(\omega-1)$ *bound relative to* $\text{rad}(abc)$, which is non-trivial when $\omega \geq 3$. It does not beat Siegel's lemma in general (since $\text{rad}(abc) \leq c$ and $1/(\omega-1) < 1$ only for $\omega \geq 3$), but it provides explicit lattice structure that was not previously formalized.

1.2 Main results

Let (a, b, c) be a coprime squarefree triple with $a + b = c$ and $\omega = \omega(abc) \geq 2$. Set $P = P_a \cup P_b \cup P_c$ and $R = \text{rad}(abc) = \prod_{p \in P} p$.

Theorem 1 (Determinant bound). $\det(F(a, b)) = R \cdot \sqrt{\sum_{p \in P} p^{-2}} < R$.

Theorem 2 (GCD lemma). *The integer coefficient vector $\mathbf{c} \in \mathbb{Z}^P$ defined by $c_p = R/p$ (for $p \in P_a \cup P_b$) and $c_p = -R/p$ (for $p \in P_c$) satisfies $\gcd(|c_p| : p \in P) = 1$.*

Corollary 3 (Vaaler lattice bound [2]). *There exists a nonzero $\psi \in F(a, b)$ with $\|\psi\|_\infty \leq \det(F(a, b))^{1/(\omega-1)} < R^{1/(\omega-1)}$.*

For non-squarefree triples, Theorem 15 below gives the upper bound $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$ for Pasten's original lattice, where $v_{\max} = \max_p v_p(abc)$. The finer exact formulas introduced for non-squarefree triples apply to a valuation-coordinate sublattice defined below, not to the original lattice without further argument.

Corollary 3 does *not* imply abc for the following reasons:

1. The bound $\|\psi\|_\infty < R^{1/(\omega-1)}$ approaches a useful bound on c only as $\omega \rightarrow \infty$. The hard cases for abc are triples with *small* ω (prime powers, Mersenne-type), where $R^{1/(\omega-1)}$ is close to $R \leq c$ —the trivial bound.
2. The vector ψ produced by Corollary 3 bounds the *lattice* minimum, not the height of c directly. The connection to abc requires the Small Derivatives Conjecture, which is equivalent to abc.
3. For the degenerate $\omega = 2$ family (e.g. $c = 2^k$, $a = 1$), the bound gives $\|\psi\|_\infty \leq R^1 = R$, which is the trivial bound. Pasten explicitly excludes this family from SDC for this reason.

2 Definitions and setup

2.1 The Pasten lattice for squarefree triples

Let (a, b, c) be a coprime squarefree triple, $a + b = c$, $a, b, c > 0$. Write $v_p(n) = 1$ for all $p \mid n$ (squarefree hypothesis).

Set $D = \text{lcm}(p : p \in P) = \prod_{p \in P} p = R$ (since P consists of distinct primes and (a, b, c) is squarefree). The additivity constraint $d^\psi(a) + d^\psi(b) = d^\psi(c)$ becomes, after multiplying by R :

$$\sum_{p \in P} c_p \cdot \psi(p) = 0,$$

where

$$c_p = \begin{cases} +R/p & p \in P_a \cup P_b, \\ -R/p & p \in P_c. \end{cases}$$

Note $c_p \in \mathbb{Z}$ since $p \mid R$. The lattice is

$$F(a, b) = L = \{\psi \in \mathbb{Z}^P \mid \mathbf{c} \cdot \psi = 0\},$$

a rank- $(\omega - 1)$ sublattice of $\mathbb{Z}^P$.

2.2 Lattice determinant

For an integer lattice defined by a single primitive constraint $\mathbf{c} \in \mathbb{Z}^\omega$ with $\text{gcd}(c_p) = 1$, the determinant equals

$$\det(L) = \|\mathbf{c}\|_2 = \left(\sum_{p \in P} c_p^2 \right)^{1/2}.$$

If $\text{gcd}(c_p) = g > 1$, the primitive form is $\mathbf{c}/g$ and $\det(L) = \|\mathbf{c}\|_2/g$.

3 Proofs

3.1 Proof of Theorem 2

Proof. Each $|c_p| = R/p = \prod_{q \in P, q \neq p} q$. Suppose a prime ℓ divides $\text{gcd}(|c_p| : p \in P)$. Since ℓ divides $|c_p|$ for every $p \in P$, in particular ℓ divides $|c_p|$ for some p ; since $|c_p| = R/p$ is a product of primes from P , we must have $\ell \in P$.

Take $p = \ell$: then $|c_\ell| = R/\ell = \prod_{q \in P, q \neq \ell} q$. This product does *not* contain ℓ as a factor, so $\ell \nmid R/\ell$. But ℓ was assumed to divide every $|c_p|$, including $|c_\ell|$. Contradiction. Hence $\text{gcd}(|c_p|) = 1$. $\square$

Remark 1. *Edge case* $|P| = 2$, $P = \{p, q\}$: $|c_p| = q$, $|c_q| = p$, $\text{gcd}(p, q) = 1$. *The case* $|P| = 1$ *is excluded by* $\omega \geq 2$.

3.2 Proof of Theorem 1

Proof. By Theorem 2, $\text{gcd}(|c_p| : p \in P) = 1$, so $\det(L) = \|\mathbf{c}\|_2$. Since $c_p = \pm R/p$ we have $c_p^2 = (R/p)^2 = R^2/p^2$, thus

$$\|\mathbf{c}\|_2^2 = \sum_{p \in P} \frac{R^2}{p^2} = R^2 \sum_{p \in P} \frac{1}{p^2}.$$

So $\det(L) = R \sqrt{\sum_{p \in P} p^{-2}}$.

It remains to show $\sum_{p \in P} p^{-2} < 1$. We prove this for any finite set P of primes by an elementary telescoping argument. Set $M = \max P$. For each $n \geq 2$:

$$\frac{1}{n^2} \leq \frac{1}{n-1} - \frac{1}{n},$$

which follows from the identity $\frac{1}{n-1} - \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} > 0$. Therefore

$$\sum_{p \in P} \frac{1}{p^2} \leq \sum_{p \in P} \left(\frac{1}{p-1} - \frac{1}{p} \right) \leq \sum_{n=2}^M \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{M} < 1.$$

(The second inequality uses $P \subseteq \{2, \dots, M\}$ and positivity of each term $1/(n-1) - 1/n > 0$.) Hence $\det(L) < R$. $\square$

Remark 2. The true value is $\sum_{p \text{ prime}} p^{-2} = P(2) \approx 0.4522$ (prime zeta function at $s = 2$). An elementary bound gives $\sum_{p \text{ prime}} p^{-2} \leq 11/18 \approx 0.611 < 1$ via $1/4 + 1/9 + \int_4^\infty x^{-2} dx = 1/4 + 1/9 + 1/4 = 11/18$.

3.3 Proof of Corollary 3

Proof. By Theorem 1, $\det(L) < R$. The lattice L has rank $\omega - 1$ as a sublattice of $\mathbb{Z}^\omega$, sitting in the hyperplane $H = \{\mathbf{x} : \mathbf{c} \cdot \mathbf{x} = 0\}$.

We use Theorem 2 of Vaaler [2], which gives a precise form of Minkowski's linear-forms theorem in the subspace setting. Let A be the $\omega \times (\omega - 1)$ matrix whose columns form a $\mathbb{Z}$ -basis of L , and take the ω ambient coordinate functionals as the linear forms. Then $\det(A^T A) = \det(L)^2$. Vaaler's theorem, with all coordinate bounds equal to $\lambda := \det(L)^{1/(\omega-1)}$, gives a nonzero $\psi \in L$ with $\|\psi\|_\infty \leq \lambda$. Equivalently, its geometric corollary says that every central hyperplane section of $[-1, 1]^\omega$ has $(\omega - 1)$ -dimensional volume at least $2^{\omega-1}$. Therefore

$$\|\psi\|_\infty \leq \det(L)^{1/(\omega-1)} < R^{1/(\omega-1)}.$$

$\square$

4 Numerical evidence

Table 1 gives comprehensive statistics from script `t30_rho_distribution.py` applied to all 44 474 squarefree coprime triples with $\omega \geq 3$ and $c \leq 1000$. Table 2 breaks out the nine universally-bounded partition types.

The unbounded types drive the high mean ρ for $\omega \leq 5$. For the bounded partition types, the distribution concentrates well below the proved upper bounds: the median ρ for combined $\omega = 4$ bounded types is 0.34, and fewer than 1% of triples come within 0.02 of $2^{-1/3}$.

ω	count	mean ρ	max ρ	% > 1	note
3	79	0.864	1.080	30.4%	includes unbounded types (0, 2, 1), etc.
4	3 223	0.945	9.919	27.7%	(2, 1, 1), (0, 1, 3), (0, 3, 1) unbounded
5	14 009	0.961	31.105	24.1%	(1, 1, 3), (1, 3, 1), (3, 1, 1) unbounded
6	18 739	0.465	17.889	5.3%	all types unbounded
7	7 766	0.248	1.631	1.0%	all types unbounded

Table 1: Empirical summary of $\rho = \|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$ organized by $\omega = \omega(abc)$ over all squarefree coprime triples with $c \leq 1000$. High mean ρ for $\omega = 3, 4, 5$ is driven by unbounded partition types.

Type	n	mean ρ	max ρ	upper bound	% > 0.5	status
(1, 1, 1), $\omega = 3$	35	0.693	0.706	$2^{-1/2} \approx 0.707$	100%	BOUNDED
(0, 2, 2), $\omega = 4$	69	0.147	0.505	$3 \cdot 210^{-1/3} \approx 0.505$	1.4%	BOUNDED
(1, 1, 2), $\omega = 4$	1 029	0.356	0.781	$2^{-1/3} \approx 0.794$	30.0%	BOUNDED
(1, 2, 1), $\omega = 4$	1 036	0.372	0.791	$2^{-1/3} \approx 0.794$	31.9%	BOUNDED
(1, 2, 2), $\omega = 5$	3 450	0.104	0.817	$2^{-1/4} \approx 0.8408$	0.0%	BOUNDED
(2, 1, 2), $\omega = 5$	3 115	0.093	0.634	$2^{-1/4} \approx 0.841$	0.3%	BOUNDED
(2, 2, 1), $\omega = 5$	3 006	0.089	0.611	$6^{-1/4} \approx 0.6389$	0.4%	BOUNDED
(0, 2, 3), $\omega = 5$	—	—	—	1	—	BOUNDED
(0, 3, 2), $\omega = 5$	—	—	—	1	—	BOUNDED

Table 2: Distribution of ρ for the nine universally-bounded partition types, $c \leq 1000$ (for $\omega \leq 4$ and $\omega = 5$ multi-group types). The displayed constants are proved upper bounds. Equality in an asymptotic bound would require an infinite prime-pattern family (such as twin primes); sharpness is therefore conditional and is not asserted by the table. P99 and P100 for combined $\omega = 4$ bounded types: 0.775 and 0.791, both below $2^{-1/3} = 0.794$.

5 Discussion

5.1 The degenerate $\omega = 2$ family

For $\omega = 2$ triples, $F(a, b)$ has rank 1, and the single generator is $\psi = (c_2/g, -c_1/g)$ where $g = \gcd(c_1, c_2)$. In the squarefree case $P = \{p, q\}$ and the minimum norm is $\max(q, p) = \max(p, q)$. For a Mersenne triple $(1, 2^k - 1, 2^k)$ with $2^k - 1$ prime: $P = \{2, 2^k - 1\}$, $R = 2(2^k - 1)$, $\|\psi\|_{\infty} = 2^k - 1 \approx R/2$. So $\|\psi\|_{\infty}/R \rightarrow 1/2$, not smaller. The bound is essentially tight.

5.2 Why this does not prove abc

Let (a, b, c) be a high-quality abc triple (quality = $\log c / \log R > 1$). High-quality triples tend to have small ω (often $\omega = 3$ or 4). Corollary 3 gives $\|\psi\|_{\infty} \leq R^{1/(\omega-1)}$. For $\omega = 3$: $\|\psi\|_{\infty} \leq R^{1/2}$. Since $R \leq c$, this is $\|\psi\|_{\infty} \leq c^{1/2}$, which is weaker than what SDC requires ($\eta < 1$) only for the case $\eta_{\text{SDC}} < 1/2$. The abc conjecture is equivalent to SDC holding for all $\varepsilon > 0$, i.e. $\|\psi\|_{\infty} = o(c^{\varepsilon})$ for all $\varepsilon > 0$ — an exponential improvement over what we prove.

The gap between our Minkowski bound ($\eta_R = 1/(\omega - 1)$) and SDC ($\eta_c = \varepsilon$ for all $\varepsilon > 0$)

is the full depth of abc .

5.3 Quality boundary for squarefree triples

For squarefree coprime (a, b, c) with $a + b = c$ and $R = \text{rad}(abc) = abc$, the quality satisfies $\text{quality} = \log c / \log(abc)$. We note a clean characterization that limits how hard abc can be within the squarefree subfamily:

Proposition 4 (Quality boundary). *For squarefree coprime (a, b, c) with $a + b = c$ and $a \leq b$:*

$$\text{quality} > \frac{1}{2} \iff a = 1.$$

Moreover, $\text{quality} < 1$ always, and $\text{quality} = \frac{1}{2}$ is never achieved. For $a = 1$, $\text{quality} \rightarrow \frac{1}{2}$ from above as $b \rightarrow \infty$. For $a = 2$, $\text{quality} \rightarrow \frac{1}{2}$ from below.

Proof. $\text{quality} > \frac{1}{2}$ iff $\log c > \log a + \log b$, i.e. $c > ab$, i.e. $a + b > ab$, i.e. $(a - 1)(b - 1) < 1$. For positive integers, this means $(a - 1)(b - 1) = 0$, i.e. $a = 1$ or $b = 1$. Since $a \leq b$ and $a \geq 1$: $b = 1 \Rightarrow a = 1$. So $\text{quality} > 1/2$ iff $a = 1$. $\text{Quality} < 1$ since $R = abc > c$. $\text{Quality} = 1/2$ would require $c = ab$, impossible since $c = a + b < ab$ for $a, b \geq 2$ and $c = a + b > ab = b$ for $a = 1$. $\square$

The Lean 4 formalization of the combinatorial core ($ab < a + b \leftrightarrow a = 1$ for $1 \leq a \leq b$) is given by `pasten.F21A.ab_lt_sum_iff_a.one` (proved by `omega/nlinarith`, zero sorry).

5.4 Quality- ρ trade-off

Proposition 4 implies that squarefree triples with $\text{quality} > 1/2$ are exactly the *trivial* triples ($a = 1$). For the harder non-trivial cases ($a \geq 2$), $\text{quality} \leq 1/2$ with equality only in the limit.

Within the sampled bounded partition types, quality and ρ point in *opposite* directions:

- **High- ρ families** (balanced types approaching the conditional extremal bound): e.g. type $(1, 1, 2)$ with $a = p \approx q$ both large odd primes. Here $\rho^3 \rightarrow 1/2$ (approaching $2^{-1/3}$) while $\text{quality} \rightarrow \log(2p) / \log(2p^3) = 1/3$.
- **High-quality families** ($a = 2$ fixed, $b \rightarrow \infty$): Here $\text{quality} \rightarrow 1/2$ (approaching the squarefree non-trivial supremum) while $\rho = 2 / (2 \cdot b \cdot (2 + b))^{1/3} \rightarrow 0$.

Empirically (Table 2, $c \leq 2000$), Pearson $r(\text{quality}, \rho) \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$.

Conclusion. The triples hardest for abc (highest quality, approaching $\text{quality} = 1/2$ from below) are not the triples closest to the extremal ρ bound in the surveyed families. The gap between “hard for abc ” and “hard for SDC” is a provable structural feature: within the squarefree subfamily, maximizing quality and maximizing ρ are *geometrically opposite* in the parameter space of partition types.

F31: joint quality- ρ bound

The opposition between quality and ρ is not merely empirical: for every *bounded- ρ* partition type (those with $\sup_{\text{triples}} \rho < \infty$) the sum $\text{quality} + \rho^{\omega-1}$ is less than 1.

Theorem 5 (F31: Quality- ρ Joint Bound). *For every squarefree coprime triple (a, b, c) with $a + b = c$, $a \geq 2$, belonging to a bounded- ρ partition type:*

$$\text{quality}(a, b, c) + \rho(a, b, c)^{\omega-1} < 1.$$

The bound is strict.

Proof. By Proposition 4 (quality boundary), $\text{quality} < 1/2$ for all non-trivial squarefree triples ($a \geq 2$). For a bounded type containing a one-prime constituent, let $g \geq 2$ be the smallest group minimum and d the second-smallest. Every remaining prime is larger than d , so $R > 2d^{\omega-1}$ and $\rho^{\omega-1} < 1/2$. The only bounded type with $a \geq 2$ and no one-prime constituent is $(2, 2, 1)$. In that case, writing g, h for the primes in the first constituent, d, q_2 for those in the second, and r for the single third-prime, with d the second-smallest group minimum, gives $ghq_2r > 6d^2$ and hence $\rho^{\omega-1} = d^3/(ghq_2r) < 1/6$. Hence $\text{quality} + \rho^{\omega-1} < 1/2 + 1/2 = 1$. $\square$

Remark 3 (Conditional tightness for type $(1, 1, 1)$). *The bound would be tight along type $(1, 1, 1)$ triples if infinitely many twin primes existed. For the twin-prime family $(2, q, q+2)$, $\text{quality} \rightarrow 1/2$ and $\rho^2 = q/(2(q+2)) \rightarrow 1/2$ as $q \rightarrow \infty$. The infinitude of twin primes is open, so this is conditional sharpness rather than an unconditional determination of the supremum. The algebraic core — $(q+2)^q < (2q)^{q+4}$ for all $q \geq 1$ (proved for $q = 1$ directly; for $q \geq 2$ by $q+2 \leq 2q$ and $q < q+4$) — is formalized as `pasten_F31a` in the Lean kernel.*

Remark 4 (Unbounded types). *The bound fails for unbounded- ρ types (e.g. type $(2, 1, 1)$: $\rho^3 \approx q/6 \rightarrow \infty$). For bounded types the gap to 1 is large away from type $(1, 1, 1)$: $\omega = 4$ bounded types $\max(\text{quality} + \rho^{\omega-1}) \approx 0.523$; $\omega = 5$ bounded types $\max(\text{quality} + \rho^{\omega-1}) \approx 0.569$ (verified numerically, T51). Under the F11 prime-pattern criterion, $\rho^{\omega-1} \rightarrow \infty$ and the bound $\text{quality} + \rho^{\omega-1} \leq 1$ fails along that sequence.*

5.5 Non-degeneracy: complete classification for $\omega \leq 5$

Corollary 3 produces a nonzero $\psi \in F(a, b)$, but the produced vector may be *degenerate*: $W^\psi(a, b) = 0$ where $W^\psi(a, b) = ab(\sum_{p \in P_b} \psi(p)/p - \sum_{p \in P_a} \psi(p)/p)$. The degenerate sublattice $L_0 = \{\psi \in F(a, b) : W^\psi = 0\}$ has rank $\omega - 2$; the smallest non-degenerate vector may be longer than the Minkowski bound. This section gives a complete analysis for all squarefree partition types with $\omega \leq 5$.

Universal Divisibility Lemma and coordinate change

Theorem 6 (F8: Universal Divisibility). *For any squarefree coprime (a, b, c) with $a + b = c$ and any $\psi \in F(a, b)$: $q \mid \psi_q$ for every prime $q \in P$.*

Proof. The integer lattice constraint isolates ψ_q : $(R/q)\psi_q = -\sum_{p \neq q} \text{sign}(p)(R/p)\psi_p$. For each $p \neq q$: R/p contains q as a factor (since $q \in P$, $q \neq p$). Hence $q \mid \text{RHS}$. Since $\gcd(q, R/q) = 1$: $q \mid \psi_q$. $\square$

Define $\varphi_q = \psi_q/q \in \mathbb{Z}$ (well-defined by Theorem 6). This gives a canonical bijection

$$F(a, b) \cong \tilde{F}(a, b) = \{\varphi \in \mathbb{Z}^\omega \mid \sum_{q \in P} \text{sign}(q) \varphi_q = 0\},$$

where $\text{sign}(q) = +1$ if $q \mid ab$ and -1 if $q \mid c$. The Wronskian becomes $W = ab(S_b - S_a)$ where $S_X = \sum_{q \mid X} \varphi_q$. Degeneracy is the *purely integer* condition $S_b = S_a$; the non-degeneracy geometry is combinatorial, determined by which primes can be “crossed” across the constraint-sign boundary.

$\omega = 3$: all non-degenerate

For squarefree prime triples (p, q, r) with $p + q = r$ (parity forces $p = 2$), the degenerate sublattice L_0 is generated by $(p, q, 2r)$ with ℓ^∞ -norm $2r$. Combining Theorem 1 with the Minkowski–Vaaler bound gives a vector of ℓ^∞ -norm at most $\sqrt{R} = \sqrt{pqr} < 2r$ (proved: $pq < 4r$ for prime triples, hence $pqr < 4r^2$).

Theorem 7 (F3: $\omega=3$ non-degeneracy). *For every squarefree prime triple (p, q, r) with $p + q = r$, the shortest non-zero vector of $F(p, q)$ is non-degenerate; equivalently, $\text{nd}(p, q) < 2r$.*

Therefore Theorem 7 says that the *shortest vector is always non-degenerate* for $\omega = 3$ prime triples. Four constituent lemmas are formalized in Lean 4 (zero sorry).

$\omega = 4$: complete partition-type classification

For squarefree coprime $\omega = 4$ triples with partition type (n_a, n_b, n_c) , $n_a + n_b + n_c = 4$, all six types are classified analytically via the φ -coordinate framework (scripts **t14–t19**, verified for all triples $c \leq 300$):

Type	Degen?	Non-degen min	Ratio bound ($\rho = \ \psi_{\text{nd}}\ /R^{1/3}$)	Pattern
(0, 2, 2)	Never	$\max(p_b^{\min}, p_c^{\min})$	$\leq 3 \cdot 210^{-1/3} \approx 0.505$, max at (1, 14, 15)	BOUNDED
(1, 1, 2)	Never	$\max(p_a, p_b)$	$< 2^{-1/3} \approx 0.794$ (F14)	BOUNDED
(1, 2, 1)	iff $a > r$	$\max(a, q)$	$< 2^{-1/3} \approx 0.794$ (F12)	BOUNDED
(2, 1, 1)	Always	$\max(p_1, r)$	grows $\sim r^{1/3}$	UNBOUNDED
(0, 1, 3)	Always	$r = b$	grows $\sim r^{2/3}$	UNBOUNDED
(0, 3, 1)	Always	$r = c$	grows $\sim r^{2/3}$	UNBOUNDED

Here r (for types (2, 1, 1), (0, 1, 3), (0, 3, 1)) denotes the single prime in the constituent with $n = 1$, which can grow without bound in infinite subfamilies. For type (1, 2, 1), the degeneracy condition $a > r$ (where r is the larger prime of b) forces $r < a \leq 2r$, and the ratio $\max(a, q)/R^{1/3}$ is bounded by $2^{-1/3}$ (proved by elementary calculus: the function $t^{2/3}/(2(t+2))^{1/3}$ is strictly increasing on (1, 2) and remains below the conditional endpoint $2^{-1/3}$ at $t \rightarrow 2$).

The *UNBOUNDED* types arise when a constituent with $n \geq 2$ prime factors shares the primes available for non-degenerate crossings with no small partner, forcing $\|\psi_{\text{nd}}\|_\infty$ to grow with the isolated prime.

$\omega = 5$: all partition types

The φ -coordinate classification extends directly to $\omega = 5$ (script `t20`, 902 triples $c \leq 200$):

Type	Degen/ND	Max ρ ($= \ \psi_{\text{nd}}\ /R^{1/4}$)	Achieved at	Pattern
(1, 2, 2)	0/219	$2^{-1/4} \approx 0.8408$ (upper bound)	—	BOUNDED (F26)
(2, 1, 2)	17/171	$2^{-1/4} \approx 0.841$ (upper bound)	—	BOUNDED (F30)
(2, 2, 1)	15/189	$6^{-1/4} \approx 0.6389$ (upper bound)	—	BOUNDED (F24)
(0, 2, 3)	—	< 1	—	BOUNDED (F23)
(0, 3, 2)	—	< 1	—	BOUNDED (F23)
(1, 1, 3)	112/118	2.798	—	UNBOUNDED (F11)
(1, 3, 1)	106/110	2.798	—	UNBOUNDED (F11)
(3, 1, 1)	0/123	6.745	—	UNBOUNDED (F11)
(0, 1, 4)	—	$\rightarrow \infty$	—	UNBOUNDED (F22)
(0, 4, 1)	—	$\rightarrow \infty$	—	UNBOUNDED (F22)

All five bounded $\omega = 5$ types have strict upper bounds. Their extremal constants are conditional on infinite prime-pattern families and are not asserted as unconditional suprema. Type (1, 2, 2) extends the universal bound $2^{-1/(\omega-1)}$ to $\omega = 5$: for $a = 2$, $b = pq$, $c = rs$ with all four odd primes near equal and $2 + pq = rs$, $\rho^4 = p^3/(2qrs) < 1/2$ since $q, r, s > p$ (Theorem 8). As all four primes $\rightarrow n$ in such a family: $\rho^4 \rightarrow 1/2$, giving conditional sharpness of $2^{-1/4}$. Verified: 0 violations of $\rho < 2^{-1/4}$ for $a = 2$, $b \leq 200\,000$. Types (2, 1, 2) and (1, 2, 2) share the conditional extremal constant $2^{-1/4} \approx 0.8409$: by the $a \leftrightarrow b$ symmetry of ρ (which depends only on $\{\min P_a, \min P_b, \min P_c\}$ and R), any (2, 1, 2) triple (a, b, c) gives a (1, 2, 2) triple (b, a, c) with the same ρ . Hence the conditional constant for (2, 1, 2) equals that for (1, 2, 2). The extremal family: $b = 2$ (fixed), $a = p_1 p_2$, $c = p_1 p_2 + 2 = r_1 r_2$ near-twin semiprimes; $\text{nd} = p_1$, $\rho^4 = p_1^3/(2 p_2 r_1 r_2) < 1/2$ (since $p_2 r_1 r_2 \geq p_1 p_2^2 > p_1^3$). As $p_1 \approx p_2 \approx r_1 \approx r_2 \rightarrow n$ in such a family: $\rho^4 \rightarrow 1/2$. Verified: 0 violations of $\rho < 2^{-1/4}$ for 2,475,742 triples at $c \leq 50,000$ (6 violations of the old incorrect bound $6^{-1/4}$; the old F25 claim was wrong).

Theorem 8 (F26: type (1, 2, 2) upper bound). *For any type (1, 2, 2) triple, $\rho^4 < 1/2$, so $\rho < 2^{-1/4}$. In the subfamily $a = 2$, $b = pq$, $c = rs$ (with $2 + pq = rs$, $p \leq q$, and $r \leq s$), $\rho^4 = p^3/(2qrs)$. Equality is never attained. The constant $2^{-1/4}$ would be sharp if the corresponding near-equal prime-pattern family were infinite; this sharpness is conditional and is not claimed as a theorem.*

Proof. Let $g \geq 2$ be the single-prime group's minimum and let d be the second smallest of the three group minima. The other two primes in the two-prime groups are strictly larger than their group minima, and the third group minimum is at least d . Therefore the three primes other than g and d have product strictly larger than d^3 . Hence

$$\rho^4 = \frac{d^4}{R} < \frac{d^4}{2d \cdot d^3} = \frac{1}{2}. \quad \square$$

Theorem 9 (F24: type (2, 2, 1) upper bound). *For any type (2, 2, 1) triple, $\rho^4 < 1/6$, so $\rho < 6^{-1/4} \approx 0.6389$. The bound is strict for every finite triple. It would be approached by an infinite $a = 6$ near-twin-prime family if such a family exists; that conditional sharpness*

is not claimed unconditionally. (Lean: `pasten.F29.221.rho4.lt.sixth`, which proves the general bound F29 covering all type (2, 2, 1) triples.)

Proof. Let g be the smallest group minimum and d the second-smallest; they lie in different groups. Let $h > g$ be the other prime in g 's constituent, let $q_2 > d$ be the other prime in d 's constituent, and let $r \geq d$ be the single prime in the third group. Then

$$\rho^4 = \frac{d^4}{ghdq_2r} = \frac{d^3}{ghq_2r} < \frac{1}{6}$$

because $g \geq 2$, $h \geq 3$, $q_2 \geq d + 1$, and $r \geq d$, whence $ghq_2r \geq 6d(d + 1) > 6d^2$. $\square$

Theorem 10 (F23: types (0, 2, 3) and (0, 3, 2) upper bound). *For either type, $\rho^4 < 1$. A supremum equal to 1 would require an infinite near-equal prime family and is therefore conditional.*

Proof. Here $a = 1$ and the two non-empty constituents each have at least two prime factors. Let $d = nd$ be the larger of their two group minima, and let Y be the constituent whose minimum is d . Write $Y = du$ with $u > d$; since the primes are distinct, $u \geq d + 2$. The other constituent is either $Y - 1$ or $Y + 1$, so it is at least $Y - 1$. Since $R = \text{rad}(abc) = bc = Y(Y \pm 1)$,

$$R \geq d(d + 2)(d^2 + 2d - 1) > d^4.$$

Thus $\rho^4 = d^4/R < 1$.

In the special subfamily $b = pq$ with $3 \leq p < q$ and $c = pq + 1$, this becomes $\rho^4 = p^3/(q(pq + 1))$, the formula used in the finite computations. $\square$

Type (2, 2, 1) has the strict upper bound $6^{-1/4}$ (Theorem 9); conditional sharpness is approached by the $a = 6$ family (see above). Types (0, 2, 3) and (0, 3, 2) have $a = 1$ (empty P_a), giving only two non-empty prime groups; their F10 $nd = \max(\min P_b, \min P_c)$ satisfies $\rho < 1$ always (Theorem 10), with conditional sharpness at 1. The unbounded types (0, 1, 4) and (0, 4, 1) have $\rho^4 \approx q^2 \rightarrow \infty$ analytically.

F10: optimal crossing formula (general ω)

The key structural insight from **t21** completes the classification at all ω :

Theorem 11 (F10). *For any cross-group pair $X \neq Y$ with $X, Y \in \{P_a, P_b, P_c\}$, the two-entry φ -vector with $\varphi_p = 1$ ($p \in X$) and $\varphi_q = \pm 1$ ($q \in Y$, sign chosen to satisfy $\sum \text{sign}_q \varphi_q = 0$) is always non-degenerate ($S_b \neq S_a$). Its ψ -norm is $\max(p, q)$.*

Consequently, the minimum non-degenerate norm satisfies

$$\|\psi_{nd}\|_\infty = \min_{X \neq Y} \max(\min X, \min Y) = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\},$$

where $\min \emptyset = \infty$.

This formula corrects **t20**, which computed only the canonical $P_a \times P_b$ crossing. The corrected classification of (3, 1, 1) is UNBOUNDED (not all-degenerate): non-degenerate vectors exist, but the minimum nd norm is $\min(b, c)$, which grows.

E10/E11/E_n: explicit minimum vectors and universal lower bound

Theorem 11 asserts an equality; we now give the explicit achieving vector and prove the matching lower bound, completing the characterisation.

Theorem 12 (E10). *For type (1, 1, 1) with $a = p \leq b = q \leq c = r$ all prime, $p + q = r$: the vector $\psi = (p, -q, 0)$ lies in $F(a, b)$, satisfies $W(\psi) = -2pq \neq 0$ (non-degenerate), and $\|\psi\|_\infty = q = \text{nd}$. (Lean: `pasten_E10_vec_in_lattice`, `nondeg`, `norm`.)*

Theorem 13 (E11). *For type (1, 1, 2) with $a = p$, $b = q$, $c = r_1 r_2$ (p, q, r_1, r_2 distinct primes, $p + q = r_1 r_2$, $r_1 < r_2$): the vector $\psi = (-p, 0, -r_1, 0)$ (i.e., $\varphi = (-1, 0, -1, 0)$ in the F8 φ -coordinates) lies in $F(a, b)$, has $W(\psi) = pq \neq 0$, and $\|\psi\|_\infty = \max(p, r_1) = \text{nd}$. Verified numerically against all 310 triples with $p \leq 19$, $c \leq 500$ (`t46c`); Lean: `pasten_E11_vec_in_lattice`, `nondeg`, `norm`.*

Theorem 14 (E_n: universal lower bound). *For any squarefree coprime triple (a, b, c) with $a + b = c$ and $\omega(abc) = \omega$, every non-degenerate $\psi \in F(a, b)$ satisfies*

$$\|\psi\|_\infty \geq \text{nd} = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\}.$$

Proof. By the Universal Divisibility Lemma (Theorem 6), $\psi_\ell = \ell \varphi_\ell$ with $\varphi_\ell \in \mathbb{Z}$. Suppose for contradiction that $\|\psi\|_\infty < \text{nd}$. Then $\ell |\varphi_\ell| \leq \|\psi\|_\infty < \text{nd}$, so $\varphi_\ell = 0$ whenever $\ell \geq \text{nd}$.

Write $\text{nd} = \text{second smallest of } \{m_a, m_b, m_c\}$ with $m_X = \min P_X$ (where $\min \emptyset = \infty$). WLOG $m_a = \min(m_a, m_b, m_c)$; then $\text{nd} = \min(m_b, m_c)$. Every $\ell \in P_b$ satisfies $\ell \geq m_b \geq \text{nd}$, so $\varphi_\ell = 0$; likewise every $\ell \in P_c$ satisfies $\ell \geq m_c \geq \text{nd}$, so $\varphi_\ell = 0$. Hence $S_b := \sum_{\ell \in P_b} \varphi_\ell = 0$ and $S_c := \sum_{\ell \in P_c} \varphi_\ell = 0$. The lattice constraint (F8 φ -form) gives $S_a + S_b = S_c$, so $S_a = 0$. Non-degeneracy requires $S_b \neq S_a$, i.e. $0 \neq 0$ —contradiction. $\square$

If m_b is the global minimum, then $\text{nd} = \min(m_a, m_c)$; every prime in $P_a \cup P_c$ is at least nd , so φ vanishes on $P_a \cup P_c$. The constraint gives $S_b = 0$, contradicting $S_b \neq S_a = 0$. The case m_c global minimum is identical with S_c in place of S_b . $\square$

Theorems 11 and 14 together give the complete characterisation: $\min_{\psi \text{ nd}} \|\psi\|_\infty = \text{nd}$, achieved and tight. Key-lemma Lean formalizations: `pasten_E10_lb_key` (Theorem 12) and `pasten_E11_lb_key` (Theorem 13). Numerical confirmation across all types: $\omega = 4$ (`t47`), $\omega = 5$ (`t48`), $\omega = 6$ (`t49`)—all verified in ≤ 1 s.

OB-13B: non-squarefree upper bound

The squarefree upper bound extends to non-squarefree triples with a $v_{\max}$ factor, where $v_{\max} = \max_{p \in P} v_p(abc)$. For the finer non-squarefree formulas, and for the remainder of this non-squarefree discussion until the next squarefree classification theorem, we use the *valuation-coordinate lattice*

$$F_{\text{val}}(a, b) = \{\psi \in F(a, b) : p \mid \psi_p \text{ for every } p \in P\}.$$

Writing $\psi_p = p \varphi_p$, its constraint is

$$\sum_{p|a} v_p(a) \varphi_p + \sum_{p|b} v_p(b) \varphi_p - \sum_{p|c} v_p(c) \varphi_p = 0, \quad (1)$$

and its Wronskian condition is $\sum_{p|b} \varphi_p \neq \sum_{p|a} \varphi_p$. The inherited norm is $\|\varphi\|_\infty = \max_{p \in P} p|\varphi_p|$. Let nd_{val} denote the minimum non-degenerate norm on this sublattice; the symbol nd in these non-squarefree type formulas means nd_{val} . This convention is essential: for $(1, 8, 9)$ the original lattice contains the non-degenerate vector $\psi = (1, 2)$ of norm 2, whereas the valuation-coordinate minimum is 9. Thus exact valuation-lattice formulas do not automatically determine the original Pasten minimum.

Theorem 15 (OB-13B: non-squarefree E_n upper bound). *For every coprime triple $a+b=c$ (not necessarily squarefree) with $c \leq 10^{12}$, write $\pi_1 < \pi_2 < \dots < \pi_{\omega^*}$ for the primes dividing R .*

$$\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}.$$

For $c > 10^{12}$ the bound also holds in all cases except possibly $a = 1$, $b = M^v$ with $M \in \{6, 10\}$, v even, $v \not\equiv 6 \pmod{12}$, or $M \geq 14$ and $\pi_1, \pi_2 \in P_b$; for those sub-cases the proof is conditional on non-existence of $R < c$ triples (see proof below).

Proof. For each cross-group prime pair (p, q) with $p \in P_g$, $q \in P_{g'}$ ($g \neq g'$) and $g_0 = \gcd(v_p(g\text{-factor}), v_q(g'\text{-factor}))$, define the vector $\varphi^{(p,q)}$ by $\varphi_p = v_q/g_0$, $\varphi_q = -v_p/g_0$, and $\varphi_r = 0$ for all $r \notin \{p, q\}$. A direct computation shows $\varphi^{(p,q)} \in F_{\text{val}}(a, b)$ (equation (1) is satisfied). The corresponding original-lattice vector $\psi^{(p,q)}$ has $\psi_p = p\varphi_p$ and $\psi_q = q\varphi_q$, and its valuation-coordinate Wronskian is $\pm v_q/g_0 \neq 0$, so it is non-degenerate. Its norm is $\|\varphi^{(p,q)}\| = \max(p \cdot v_q/g_0, q \cdot v_p/g_0) \leq v_{\max} \cdot \max(p, q)$.

Taking the minimum over all cross-group pairs and noting that when the optimal pair uses primes from groups G_1, G_2 , we have $\max(\min G_1, \min G_2) \leq \text{med}(m_a, m_b, m_c)$:

$$\text{nd}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c).$$

It remains to show $\pi_* := \text{med}(m_a, m_b, m_c) \leq R^{1/(\omega^*-1)}$, i.e. $\pi_*^{\omega^*-1} \leq R$. We split into two cases.

Case 1. The two globally smallest primes $\pi_1 < \pi_2$ in P lie in *different* groups (or $P_a \neq \emptyset$ and a within-group vector on P_a or P_b uses primes with unequal valuations, giving a vector of norm $\leq v_{\max} \cdot \pi_2$). Then $\pi_* \leq \pi_2$. The $\omega^* - 1$ primes of R other than π_1 all satisfy $q_i \geq \pi_2$, so their product $\geq \pi_2^{\omega^*-1}$, giving $R \geq \pi_1 \cdot \pi_2^{\omega^*-1} \geq \pi_2^{\omega^*}$, i.e. $\pi_2 \leq R^{1/(\omega^*-1)}$. ✓

Case 2. $P_a = \emptyset$, P_b has equal valuations (every $p \in P_b$ satisfies $v_p(b) = v$), and $\pi_1, \pi_2 \in P_b$. Then within-group on P_b is degenerate ($W = 0$), and the optimal P_b - P_c cross-group pair gives $\pi_* = \min(P_c) = \pi_3$ (the third-smallest prime globally). We need $\pi_3^{\omega^*-1} \leq R$. Write $R/\pi_3 = \pi_1 \cdot \pi_2 \cdot \prod_{j \geq 4} \pi_j \geq 6 \cdot (\pi_3 + 2)^{\omega^*-3}$ (using $\pi_1 = 2$, $\pi_2 = 3$, all $\pi_j \geq \pi_3 + 2$ for $j \geq 4$; there are $\omega^* - 3$ such primes). For $\pi_3 \leq 7$ and $\omega^* \geq 4$: we need $\pi_3^{\omega^*-2} \leq 6 \cdot (\pi_3 + 2)^{\omega^*-3}$. For $\pi_3 = 7$, $\omega^* = 4$: $7^2 = 49 \leq 6 \cdot 9 = 54$. ✓ The ratio $7^{\omega^*-2}/(6 \cdot 9^{\omega^*-3}) = (49/54) \cdot (7/9)^{\omega^*-4} \leq 49/54 < 1$ for all $\omega^* \geq 4$. ✓ (For $\pi_3 = 5$: $5^2 = 25 \leq 6 \cdot 7 = 42$, ratio $(25/42) \cdot (5/7)^{\omega^*-4} < 1$. ✓) For $\pi_3 \geq 11$: we analyse the two families of arising triples.

Case 2a: $M = 10$, v odd ($P_b = \{2, 5\}$, $k = 2$). Since $10 \equiv -1 \pmod{11}$, we have $11 \mid 1 + 10^v$ for all odd v , so $\pi_3 = 11$. Then $R/\pi_3 = 10 \cdot \prod_{j \geq 4} \pi_j \geq 10 \cdot 13^{\omega^*-3}$. For $\omega^* = 4$: $10 \cdot 13 = 130 \geq 121 = 11^2$. ✓ The ratio $10 \cdot 13^{\omega^*-3}/11^{\omega^*-2} = (130/121) \cdot (13/11)^{\omega^*-4} > 1$ for all $\omega^* \geq 4$. ✓

Case 2b: $M = 6$, $v \equiv 6 \pmod{12}$ ($P_b = \{2, 3\}$, $k = 2$). Since $5 \nmid c$ (as $6 \equiv 1 \pmod{5}$), $7 \nmid c$ (v even, $6 \equiv -1 \pmod{7}$), and $11 \nmid c$ ($\text{ord}(6 \pmod{11}) = 10$, $6^v \equiv -1 \pmod{11}$) requires

$v \equiv 5 \pmod{10}$, which is odd), we have $\pi_3 \geq 13$. For $\pi_3 = 13$ (requiring $v \equiv 6 \pmod{12}$): the cyclotomic factor $\Phi_{12}(6) = 6^4 - 6^2 + 1 = 1261 = 13 \times 97$ shows $\text{ord}(6 \pmod{13}) = \text{ord}(6 \pmod{97}) = 12$ and $6^6 \equiv -1$ modulo both 13 and 97. Moreover $\text{ord}(6 \pmod{37}) = 4$ with $6^2 \equiv -1 \pmod{37}$, and $v \equiv 6 \pmod{12}$ implies $v \equiv 2 \pmod{4}$, so $37 \mid c$ too. Hence $P_c \supseteq \{13, 37, 97\}$, $|P_c| \geq 3$, $\omega^* \geq 5$, $R \geq 6 \cdot 13 \cdot 37 \cdot 97 = 279\,942$. For $\omega^* = 5$: $\pi_3^{\omega^*-1} = 13^4 = 28\,561 \leq 279\,942 = R$. ✓ For $\omega^* > 5$: each extra P_c prime satisfies $q \geq \pi_3 + 2 = 15$, so $R/\pi_3^{\omega^*-2}$ grows by a factor $q/13 \geq 15/13 > 1$ per additional prime. ✓

For all other even- v sub-cases ($M = 6$, v even, $v \not\equiv 6 \pmod{12}$; $M = 10$, v even; $M \geq 14$): exhaustive search (t96) over $c \leq 10^{12}$ finds no $R < c$ triple with $\pi_1, \pi_2 \in P_b$ and $|P_c| \geq 2$. (A complete unconditional proof of non-existence in these sub-cases — showing that $1 + M^v$ is squarefree enough to force $R \geq c$ — requires a Wieferich-type prime-power analysis and is an open auxiliary obligation. The theorem therefore holds unconditionally for $c \leq 10^{12}$ in these sub-cases and is conditional on non-existence of such triples for larger c .)

Verification summary: the complete list of $R < c$ edge-case triples with $\pi_1, \pi_2 \in P_b$ and $c \leq 10^{12}$ is $(1, 6^7, 279937)$ ($\pi_3 = 7$, proved above) and $(1, 10^{11}, 10^{11} + 1)$ ($\pi_3 = 11$, Case 2a); both satisfy $\pi_3^{\omega^*-1} \leq R$ with margins $\times 100$ and $\times 565\,000$ respectively. ✓ $\square$

Remarks. (i) For squarefree triples $v_{\max} = 1$ and this recovers the Minkowski bound. (ii) The two-prime vector with GCD reduction can be sharpened to a multi-prime “balanced load” construction: for triples $(1, b, q^k)$ with b squarefree and $|P_b| \geq 2$, the optimal vector distributes the constraint k across a subset $T \subseteq P_b$, achieving norm $\max(q, k / \sum_{p_i \in T} p_i^{-1})$ — strictly below the pairwise bound. The exact formula $\text{nd}_{\text{val}}(1, b) = \max(q, \min_T \lceil k/H(T) \rceil^*)$ with $H(T) = \sum_{p_i \in T} 1/p_i$ is verified for all tested cases; no simple closed form exists for arbitrary valuation profiles, but $\text{nd}_{\text{val}}(a, b)$ is computable exactly via LLL lattice reduction (Theorem 17 below, OB-13C). (iii) Numerical verification (t59, t65, t96): OB-13B holds for all 24 high-quality triples with quality > 1.15 and $c \leq 5000$, with $\text{nd}/(v_{\max} R^{1/(\omega^*-1)}) \leq 0.50$ in all cases. The tightest example is $(1, 8, 9)$: $\text{nd} = 9$, $v_{\max} = 3$, $R = 6$, ratio = 0.50. Case 2 edge-case examples: $(1, 6^7, 279937)$ with $P_b = \{2, 3\}$, $P_c = \{7, 29, 197\}$, $\pi_3 = 7$, $R = 239946$, norm = $49 \leq 154.9 = v_{\max} R^{1/4}$ (ratio 0.32); and $(1, 10^{11}, 10^{11} + 1)$ with $P_b = \{2, 5\}$, $P_c = \{11, 23, 4093, 8779\}$, $\pi_3 = 11$, $R = 90909090910$, norm = $121 \leq 1710.5$ (ratio 0.07). (iv) *Within-group construction.* When a group P_g contains two primes p, q with *distinct* p -adic valuations $v_p \neq v_q$ of the g -factor, a within-group vector $\varphi_p = v_q/g_0$, $\varphi_q = -v_p/g_0$ (all other coordinates zero, $g_0 = \gcd(v_p, v_q)$) satisfies the constraint (1) (both sides of 1 vanish for this vector) and has $W(\varphi) = \pm(v_q - v_p)/g_0 \neq 0$. This construction can give $\text{nd}_{\text{val}}(a, b) < \text{med}(m_a, m_b, m_c)$, i.e. below the squarefree E_n formula. Example: $(a, b, c) = (72, 11, 83)$ with $a = 2^3 \cdot 3^2$, $P_a = \{2, 3\}$, $P_b = \{11\}$, $P_c = \{83\}$. The within-group vector $\varphi_2 = 2$, $\varphi_3 = -3$ satisfies $3 \cdot 2 + 2 \cdot (-3) = 0$ and $W = 1 \neq 0$, giving $\text{nd}_{\text{val}}(72, 11) \leq 9 < 11 = \text{med}(m_a, m_b, m_c)$. Thus the squarefree E_n lower bound $\text{nd} \geq \text{med}$ does *not* extend to non-squarefree triples when groups have ≥ 2 primes with distinct valuations.

Corollary 16 (OB-13B sharp for the valuation lattice). *Under the same hypotheses,*

$$\text{nd}_{\text{val}}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c),$$

where med denotes the second smallest of $\{m_a, m_b, m_c\}$ (with $m_g = \infty$ for empty groups).

This bound is tight: equality holds at $(a, b, c) = (1, 8, 9)$, where $v_{\max} = 3$, $\text{med} = 3$, and $\text{nd}_{\text{val}} = 9$.

Proof. The proof of Theorem 15 constructs $\text{nd}_{\text{val}}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c)$ as an intermediate step (the Minkowski step that follows gives only a weaker bound). For $(1, 8, 9)$: $P_a = \emptyset$, $P_b = \{2\}$, $P_c = \{3\}$; $v_{\max} = 3$ (from $v_2(8) = 3$), $\text{med}(\infty, 2, 3) = 3$. The optimal vector $\varphi_2 = 2$, $\varphi_3 = 3$ satisfies $3 \cdot 2 = 2 \cdot 3$ (constraint) with $W = 2 \neq 0$ and $\text{norm} \max(2 \cdot 2, 3 \cdot 3) = 9 = 3 \cdot 3$. $\square$

OB-13C: algorithmic exact computation of $\text{nd}_{\text{val}}(a, b)$

While no simple closed-form formula generalises the squarefree E_n expression $\text{nd}_{\text{sq}} = \text{med}(m_a, m_b, m_c)$ to all non-squarefree triples, the exact value of $\text{nd}_{\text{val}}(a, b)$ is computable by a finite deterministic lattice-reduction and enumeration procedure.

Theorem 17 (OB-13C: valuation-lattice SVP algorithm). *Let $a + b = c$ be a coprime triple with $\omega^* = |P|$ distinct prime factors. Assume $\omega^* \geq 2$; the unique $\omega^* = 1$ triple $(1, 1, 2)$ has rank 0 and no nonzero lattice vector. Write the constraint (1) as $\sum_{i=1}^{\omega^*} \alpha_i \varphi_i = 0$ where $\alpha_i = v_{p_i}(a)$ for $p_i \in P_a$, $\alpha_i = v_{p_i}(b)$ for $p_i \in P_b$, and $\alpha_i = -v_{p_i}(c)$ for $p_i \in P_c$. The solution lattice $L = \{\varphi \in \mathbb{Z}^{\omega^*} : \sum \alpha_i \varphi_i = 0\}$ has rank $\omega^* - 1$.*

1. *A $\mathbb{Z}$ -basis for L is computed in $O(\omega^{*2})$ operations by repeated application of the extended Euclidean algorithm to the coefficient vector (α_i) .*
2. *Applying the LLL algorithm to this basis with the weighted inner product*

$$\langle u, v \rangle = \sum_i p_i^2 u_i v_i$$

yields a reduced basis $\{b_1, \dots, b_{\omega^-1}\}$ with Gram-Schmidt vectors $b_1^*, \dots, b_{\omega^*-1}^*$.*

3. *Set $U := v_{\max} \cdot \text{med}(m_a, m_b, m_c)$, where $m_g = \min P_g$ and empty group minima are interpreted as ∞ . This is the unconditional two-prime witness bound constructed in the proof of Theorem 15. A standard depth-first enumeration over coordinates $c_{\omega^*-1}, \dots, c_1$ with half-width*

$$R_j := \left\lceil \frac{U}{\|b_j^*\|_W} \right\rceil$$

at level j (center updated from outer coordinates at each level; see proof) recovers the shortest non-degenerate valuation-coordinate vector, hence $\text{nd}_{\text{val}}(a, b)$ exactly. The values R_j are computable after Step 2. A uniform bound is $R_j \leq 2^{(\omega^-2)/2} U / p_1$ where $p_1 = \min P$.*

Proof. The algorithm is implemented in `t64_omega4_lattice_svp.py` and checked against brute-force enumeration: $\omega^* = 3$: 13 test cases, all exact (`t63`); $\omega^* = 4$: 13 test cases, all exact (`t64`); $\omega^* = 5$: 6 test cases, all exact (`t64`).

Step 1: standard Smith normal form. The unimodular column transformation reducing $[\alpha_1, \dots, \alpha_{\omega^*}]$ to $[d, 0, \dots, 0]$ ($d = \gcd(\alpha_i)$) supplies a $\mathbb{Z}$ -basis for L as its last $\omega^* - 1$ columns;

step 2 is standard LLL theory. The non-degeneracy constraint ($W \neq 0$) eliminates the degenerate sublattice $D \subset L$.

Step 3: let $\varphi^* = \sum_j c_j b_j$ be the nd-minimiser. The two-prime valuation-coordinate witness gives $\|\varphi^*\|_W \leq \text{nd}_{\text{val}}(a, b) \leq U$. Expanding in the Gram–Schmidt basis: $\varphi^* = \sum_j \hat{c}_j b_j^*$ where $\hat{c}_j = c_j + \sum_{i>j} c_i \mu_{i,j}$ (the GS coefficient). Cauchy–Schwarz in the weighted inner product gives $|\hat{c}_j| \leq \|\varphi^*\|_W / \|b_j^*\|_W \leq U / \|b_j^*\|_W$. Hence $c_j = \hat{c}_j - T_j$ where $T_j := \sum_{i>j} c_i \mu_{i,j}$ is determined by the outer coordinates $c_{j+1}, \dots, c_{\omega^*-1}$, and $|c_j - (-T_j)| \leq R_j$. The enumeration proceeds outermost-first ($j = \omega^*-1$ down to 1); at level j , with all c_i ($i > j$) fixed, the center $-T_j$ is computed and c_j is searched in $[-T_j - R_j, -T_j + R_j]$. For $j = \omega^*-1$ (outermost), $T_{\omega^*-1} = 0$, so the range is $[-R_{\omega^*-1}, R_{\omega^*-1}]$. Since $\|b_j^*\|_W^2 \geq (1/2)^{j-1} \|b_1\|_W^2 \geq (1/2)^{j-1} p_1^2$ (Lovász condition with $\delta = 3/4$; every non-zero L -vector has $\|\cdot\|_W \geq p_1$), the uniform bound $R_j \leq 2^{(j-1)/2} U / p_1 \leq 2^{(\omega^*-2)/2} U / p_1$ follows. $\square$

Universal lower bound and the $\omega^* = 2$ exact formula

The cross-group upper bound (Theorem 15) is complemented by a universal lower bound that holds for all coprime triples, squarefree or not.

Lemma 18 (Single-prime constraint). *Let $p \in P$ be any prime in $\text{rad}(abc)$. If $\varphi \in F_{\text{val}}(a, b)$ satisfies $\varphi_q = 0$ for all $q \in P \setminus \{p\}$, then $\varphi_p = 0$.*

Proof. Each prime lies in exactly one of P_a, P_b, P_c . With $\varphi_q = 0$ for all $q \neq p$, constraint (1) reduces to $v_p(\cdot) \varphi_p = 0$, where $v_p(\cdot) \geq 1$ since $p \in P$. Hence $\varphi_p = 0$. $\square$

Theorem 19 (Universal lower bound for nd_{val}). *For every coprime triple $a + b = c$, let $p_1 < p_2$ be the two smallest primes in $P = \text{rad}(abc)$. Then every non-degenerate vector in $F_{\text{val}}(a, b)$ has norm at least*

$$p_2.$$

Proof. Let $\varphi \in F_{\text{val}}(a, b)$ with $W(\varphi) \neq 0$. Suppose $\|\varphi\| < p_2$. For any $p \geq p_2$ in P , $p |\varphi_p| \leq \|\varphi\| < p_2 \leq p$ forces $|\varphi_p| < 1$, so $\varphi_p = 0$. The only prime in P with $p < p_2$ is p_1 , so $\varphi_q = 0$ for all $q \neq p_1$. By Lemma 18, $\varphi_{p_1} = 0$ as well, giving $\varphi = 0$, which contradicts $W(\varphi) \neq 0$. $\square$

Remark 5. *The bound is tight: for $(a, b, c) = (1, 2, 3)$ with $P = \{2, 3\}$, the vector $\varphi_2 = \varphi_3 = 1$ satisfies the constraint ($1 \cdot 1 = 1 \cdot 1$) and has $W = \varphi_2 = 1 \neq 0$, giving $\text{nd}_{\text{val}}(1, 2) = 3 = p_2$.*

For squarefree triples the E_n theorem gives $\text{nd}_{\text{val}}(a, b) = \text{med}(m_a, m_b, m_c) \geq p_2$ (each group minimum is a prime in P), so Theorem 19 is subsumed by E_n in that case. For non-squarefree triples it is the tightest universal lower bound: the within-group construction of Remark (iv) shows that $\text{nd}_{\text{val}}(a, b)$ can fall strictly below $\text{med}(m_a, m_b, m_c)$ (counterexample (72, 11, 83): $\text{nd}_{\text{val}} = 9 > p_2 = 3$), so Theorem 19 is the right replacement for E_n 's lower bound.

Verified on 10 named non-squarefree examples including the within-group counterexample (72, 11, 83) (t67); the analytical proof above covers all triples.

For triples with $\omega^* = 2$ the lower bound is sharp and the exact value is determined by a one-dimensional lattice.

Theorem 20 (Exact nd_{val} for $\omega^* = 2$). *Let $a + b = c$ be a coprime triple with $\omega^* = 2$, so $P = \{p, q\}$ with $p < q$. Let v be the p -adic valuation and w the q -adic valuation of their respective factors in abc , and set $g = \gcd(v, w)$. Then*

$$\text{nd}_{\text{val}}(a, b) = \max\left(p \cdot \frac{w}{g}, q \cdot \frac{v}{g}\right) \geq q = p_2.$$

Equality $\text{nd}_{\text{val}} = q$ holds if and only if $v = w = 1$.

Proof. The constraint (1) is a single equation $v \varphi_p = w \varphi_q$ (with appropriate sign depending on the group memberships of p and q). The solution lattice L has rank 1; its primitive generator is $(\varphi_p, \varphi_q) = (w/g, \pm v/g)$. All solutions are $k \cdot (w/g, \pm v/g)$ for $k \in \mathbb{Z}$. Since $W(\varphi) \neq 0$ iff the P_b -component is non-zero, this requires $k \neq 0$. The norm at $|k| = 1$ is $\max(p \cdot w/g, q \cdot v/g)$, which is the minimum since the norm scales as $|k|$. The inequality $\geq q$ follows from $q \cdot v/g \geq q$ (as $v/g \geq 1$) and $p \cdot w/g \geq p \geq 2$. Equality $\text{nd}_{\text{val}} = q$ requires $p \cdot w/g \leq q$ and $q \cdot v/g = q$, i.e. $v = g$; combined with $w/g = 1$ from the bound $p \cdot w/g \leq q$ this gives $v = w = g = 1$. $\square$

Corollary 21 (Explicit $\omega^* = 2$ formula). *Under the hypotheses of Theorem 20, with $v = v_{p_1}$ and $w = v_{p_2}$:*

$$\text{nd}_{\text{val}}(a, b) = \frac{\max(p_1 \cdot v_{p_2}, p_2 \cdot v_{p_1})}{\gcd(v_{p_1}, v_{p_2})}.$$

In particular, $p_2 \leq \text{nd}_{\text{val}}(a, b) \leq p_2 \cdot v_{\max}$ with $v_{\max} = \max(v_{p_1}, v_{p_2})$.

Proof. Substituting $p = p_1$, $q = p_2$, $v = v_{p_1}$, $w = v_{p_2}$, $g = \gcd(v_{p_1}, v_{p_2})$ in Theorem 20 gives $\text{nd}_{\text{val}}(a, b) = \max(p_1 w, p_2 v)/g = \max(p_1 v_{p_2}, p_2 v_{p_1})/\gcd(v_{p_1}, v_{p_2})$ directly. The lower bound $\text{nd}_{\text{val}} \geq p_2$ follows from $p_2 v/g \geq p_2 \geq p_2$ (as $v \geq g$). The upper bound $\text{nd}_{\text{val}} \leq p_2 v_{\max}$ follows from $\max(p_1 w, p_2 v)/g \leq \max(p_2 w, p_2 v)/g = p_2 \cdot \max(v, w)/g \leq p_2 v_{\max}$ (since $p_1 \leq p_2$ and $g \geq 1$). $\square$

Corollary 22 (Complete norm spectrum for $\omega^* = 2$). *Under the hypotheses of Theorem 20, the set of achievable norms of non-degenerate vectors in $F(a, b)$ is exactly*

$$\Sigma(a, b) = \{k \cdot \text{nd}_{\text{val}}(a, b) : k \in \mathbb{Z}_{>0}\}.$$

In particular, the successive minima satisfy $\lambda_k = k \cdot \text{nd}_{\text{val}}(a, b)$ for all $k \geq 1$.

Proof. The proof of Theorem 20 shows that all non-degenerate lattice vectors are $m \cdot (w/g, \pm v/g)$ for $m \in \mathbb{Z} \setminus \{0\}$, with norm $|m| \cdot \max(pw/g, qv/g) = |m| \cdot \text{nd}_{\text{val}}(a, b)$. Thus every positive multiple of nd_{val} is achieved (take $m = k$), and no other values occur. $\square$

Theorem 23 (Exact nd_{val} for type $(k, 1, 1)$ non-squarefree triples). *Let $a + b = c$ be a coprime triple with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ (one prime per group, $p < q < r$) and $v_p(a) = k \geq 1$, $v_q(b) = v_r(c) = 1$. Then*

$$\text{nd}_{\text{val}}(a, b) = \min(r, qk).$$

Proof. Upper bound. The vector $\varphi_p = 0, \varphi_q = \varphi_r = 1$ satisfies the constraint ($k \cdot 0 + 1 = 1$) and has $W = 1 - 0 = 1 \neq 0$, giving norm r . The vector $\varphi_p = -1, \varphi_q = k, \varphi_r = 0$ satisfies the constraint ($k(-1) + k = 0$), has $W = k - (-1) = k + 1 \neq 0$, and norm $\max(p, qk) = qk$ (as $q > p$ and $k \geq 1$ give $qk \geq q > p$). Hence $\text{nd}_{\text{val}}(a, b) \leq \min(r, qk)$.

Lower bound. Let $\varphi = (\varphi_p, \varphi_q, \varphi_r)$ be a non-degenerate vector.

- If $\varphi_r \neq 0$: then $r|\varphi_r| \geq r$, so $\|\varphi\| \geq r \geq \min(r, qk)$.
- If $\varphi_r = 0$: the constraint reduces to $k\varphi_p + \varphi_q = 0$, i.e. $\varphi_q = -k\varphi_p$. Non-degeneracy requires $\varphi_q \neq \varphi_p$, i.e. $(k+1)\varphi_p \neq 0$, so $\varphi_p \neq 0$. Then $\|\varphi\| = \max(p|\varphi_p|, qk|\varphi_p|) = qk|\varphi_p| \geq qk \geq \min(r, qk)$.

In both cases $\|\varphi\| \geq \min(r, qk)$. $\square$

Remark 6. For $k = 1$ (squarefree), $\min(r, q) = q$ (as $q < r$), recovering the E_n value. For $k \geq 2$: $\text{nd}_{\text{val}} = qk$ when $qk < r$ (the within-group effective valuation dominates), and $\text{nd}_{\text{val}} = r$ otherwise. The crossover $qk = r$ gives a transition between two regimes.

Examples. (4, 3, 7): $p = 2, q = 3, r = 7, k = 2$. $\min(7, 6) = 6 = qk < r$ (dominant: valuation regime). (4, 7, 11): $p = 2, q = 7, r = 11, k = 2$. $\min(11, 14) = 11 = r < qk$ (dominant: pairwise regime).

This gives a tight two-sided bound: $\min(r, qk) \geq q = p_2$ (confirming Theorem 19), since $qk \geq q$ and $r > q$.

Theorem 24 (F34: Second successive minimum for type $(k, 1, 1)$). *Under the hypotheses of Theorem 23, the second successive minimum is:*

(i) *Valuation regime* ($qk < r, \text{nd}_{\text{val}} = qk$):

$$\lambda_2 = \min(r, 2qk).$$

(ii) *Pairwise regime* ($qk \geq r, \text{nd}_{\text{val}} = r$):

$$\lambda_2 = \min\{N \in \{2r, qk, q(k-1)\} : N > r\}.$$

Proof. Achievability. We exhibit a non-degenerate vector for each candidate:

- $2r$: $\varphi = (0, 2, 2)$ (i.e. $\varphi_p = 0, \varphi_q = \varphi_r = 2$). Constraint: $k \cdot 0 + 2 = 2\checkmark$. $W = 2 - 0 = 2 \neq 0$. Norm = $\max(0, 2q, 2r) = 2r$.
- qk : $\varphi = (1, -k, 0)$. Constraint: $k - k = 0\checkmark$. $W = -k - 1 \neq 0$. Norm = $\max(p, qk, 0) = qk$.
- $q(k-1)$ (when $q(k-1) > r$): $\varphi = (1, 1-k, 1)$. Constraint: $k + (1-k) = 1\checkmark$. $W = (1-k) - 1 = -k \neq 0$. Norm = $\max(p, q(k-1), r) = q(k-1)$ (since $q(k-1) > r > p$).

Gap (no norm between nd_{val} and λ_2). Let $M = \lambda_2$ and suppose $\varphi \neq 0$ is non-degenerate with $\text{nd}_{\text{val}} < \|\varphi\| < M$.

(i) *Valuation regime* ($\text{nd}_{\text{val}} = qk, M = \min(r, 2qk)$): $r|\varphi_r| \leq \|\varphi\| < M \leq r$ forces $\varphi_r = 0$. Then $k\varphi_p + \varphi_q = 0$, so $\|\varphi\| = qk|\varphi_p| \geq qk = \text{nd}_{\text{val}}$. Equality gives $|\varphi_p| = 1$; the next value is $2qk \geq M$.

(ii) *Pairwise regime* ($\text{nd}_{\text{val}} = r$): write $\varphi_p = t, \varphi_q = j - kt$ for integer t , where $\varphi_r = j$ (any integer).

- $|j| = 0$: $\|\varphi\| = qk|t| \geq qk \geq M$.
- $|j| \geq 2$: $\|\varphi\| \geq 2r \geq M$.
- $|j| = 1$ (say $j = 1$): $\|\varphi\| = \max(p|t|, q|1 - kt|, r)$. For $\|\varphi\| < M$, we need $q|1 - kt| < M$. Since $M \leq q(k - 1)$ (or $M = \min(2r, qk)$ when $q(k - 1) \leq r$), one checks that $|1 - kt| \leq k - 2$ (or $|1 - kt| \leq k - 1$ resp.) forces $kt \in [3 - k, k - 1]$, leaving only $t = 0$ (giving $\|\varphi\| = r = \text{nd}_{\text{val}}$) as an integer solution.

In every case $\|\varphi\| \geq M$, so no norm lies strictly in $(\text{nd}_{\text{val}}, M)$. $\square$

Remark 7. *The formula specialises cleanly: for $k = 1$ (type $(1, 1, 1)$, always valuation), $\lambda_2 = \min(r, 2q) = r$ since $r = q + 2 < 2q$ for twin primes $q \geq 3$ (recovering Theorem 31). For $k = 2$ pairwise ($r \leq 2q$): since $q(k - 1) = q \leq r$, one has $\lambda_2 = \min(2r, 2q)$. T87 verifies Theorem 24 on 55 triples (four base primes $p \in \{2, 3, 5, 7\}$, $k \leq 9$, adaptive bound), zero failures.*

Theorem 25 (Exact nd_{val} for type $(k, m, 1)$ non-squarefree triples). *Let $a + b = c$ be a coprime triple with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ and $v_p(a) = k \geq 1$, $v_q(b) = m \geq 1$, $v_r(c) = 1$, $p < q < r$. Let $g = \gcd(k, m)$. Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(r, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

Proof. Upper bound. The vector $\varphi_p = -m/g$, $\varphi_q = k/g$, $\varphi_r = 0$ satisfies the constraint $(k \cdot (-m/g) + m \cdot (k/g) = 0)$, has $W = k/g + m/g = (k + m)/g \neq 0$, and norm $\max(pm/g, qk/g)$. Hence $\text{nd}_{\text{val}}(a, b) \leq \max(pm/g, qk/g)$. When $\max(pm/g, qk/g) \leq r$ this gives the desired upper bound $\text{nd}_{\text{val}}(a, b) \leq \min(r, \max(pm/g, qk/g))$. When $\max(pm/g, qk/g) > r$ (pairwise regime, which forces $m = 1$ for actual triples $a = p^k$, $b = q$, $c = p^k + q$ prime), the upper bound $\text{nd}_{\text{val}}(a, b) \leq r$ follows from Theorem 23 (the vector $\varphi_p = 0$, $\varphi_q = 1$, $\varphi_r = 1$ has norm r when $m = 1$).

Lower bound — $\varphi_r = 0$ case. The constraint reduces to $k\varphi_p + m\varphi_q = 0$. Since $\gcd(m/g, k/g) = 1$, the integer solutions are $\varphi_p = -(m/g)t$, $\varphi_q = (k/g)t$ for $t \in \mathbb{Z}$. Non-degeneracy $W = (k/g + m/g)t \neq 0$ forces $t \neq 0$, giving norm $\max(pm/g, qk/g) \cdot |t| \geq \max(pm/g, qk/g)$.

Lower bound — $\varphi_r \neq 0$ case. Then $r|\varphi_r| \geq r$, so $\|\varphi\| \geq r$.

Combining both cases, $\text{nd}_{\text{val}}(a, b) \geq \min(r, \max(pm/g, qk/g))$. $\square$

Remark 8. *Theorem 23 is the special case $m = 1$ (so $g = 1$, $\max(p, qk) = qk$ since $qk > p$ for $k \geq 1$, $q > p$). When $k = m$ (equal valuations), $g = k$ and $\max(p, q) = q$, giving $\text{nd}_{\text{val}} = q = p_2$: equal valuations always realise the lower bound. When $k = m = 1$ (squarefree), $\text{nd}_{\text{val}} = q$, recovering E_n .*

Corollary 26 (Valuation regime is independent of n). *Let $a = p^k$, $b = q^m$, $c = r^n$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ and $p < q < r$. Set $g = \gcd(k, m)$ and $N_0 = \max(pm/g, qk/g)$. If $N_0 \leq r$ (valuation regime), then*

$$\text{nd}_{\text{val}}(a, b) = N_0 = \max\left(\frac{pm}{g}, \frac{qk}{g}\right),$$

regardless of the exponent $n = v_r(c)$.

Proof. The witness $\varphi_p = -m/g$, $\varphi_q = k/g$, $\varphi_r = 0$ satisfies $k(-m/g) + m(k/g) = 0 = n \cdot 0$ for any n , has $W = (k + m)/g \neq 0$, and achieves norm N_0 . The lower bound argument of Theorem 25 is verbatim: when $\varphi_r = 0$ the constraint reduces to $k\varphi_p + m\varphi_q = 0$, giving norm $\geq N_0$; when $\varphi_r \neq 0$, norm $\geq r \geq N_0$. Neither case involves n . $\square$

Theorem 27 (F35: Second successive minimum for type $(k, m, 1)$ in the valuation regime). *Let $a + b = c$ be a coprime triple with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$, $v_p(a) = k \geq 2$, $v_q(b) = m \geq 2$, $v_r(c) = 1$, $p < q < r$. Set $g = \gcd(k, m)$ and $N_0 = \max(pm/g, qk/g)$. In the valuation regime ($N_0 < r$), the second successive minimum satisfies*

$$\lambda_2(a, b) = \min(r, 2N_0),$$

with the convention that r is achievable only when $g = 1$ (coprime exponents); when $g \geq 2$, the $|\varphi_r| = 1$ branch has no integer solutions and $\lambda_2 = 2N_0$ (which equals $\min(r, 2N_0)$ in all practical cases since $2N_0 \leq r$ holds whenever $g \geq 2$ for prime triples with $k, m \geq 2$).

Proof. Upper bound. The vector $\varphi_p = -2m/g$, $\varphi_q = 2k/g$, $\varphi_r = 0$ lies in the lattice ($k(-2m/g) + m(2k/g) = 0$), has $W = 2k/g + 2m/g = 2(k + m)/g \neq 0$, and achieves norm $2N_0 > N_0 = \text{nd}_{\text{val}}(a, b)$. When $g = 1$ and $r \leq 2N_0$: the minimal Bezout solution (k_0, m_0) of $kk_0 + mm_0 = 1$ satisfies $|k_0| \leq m/2$ and $|m_0| \leq k/2$, so $p|k_0| \leq N_0/2 < r$ and $q|m_0| \leq N_0/2 < r$; thus the vector $(\varphi_p, \varphi_q, \varphi_r) = (k_0, m_0, 1)$ has norm $\max(p|k_0|, q|m_0|, r) = r$. Hence $\lambda_2 \leq \min(r, 2N_0)$.

Lower bound. Split every non-minimizing vector by its φ_r value:

- $\varphi_r = 0$: the constraint reduces to $k\varphi_p + m\varphi_q = 0$, whose integer solutions are $(\varphi_p, \varphi_q) = (-(m/g)t, (k/g)t)$ for $t \in \mathbb{Z}$. Since $\gcd(m/g, k/g) = 1$, the generator is primitive, and norm $= |t|N_0 \geq 2N_0$ for $|t| \geq 2$.
- $|\varphi_r| \geq 1$: norm $\geq r \cdot |\varphi_r| \geq r \geq \min(r, 2N_0)$ (using $N_0 < r$).

Every non-minimizing vector therefore has norm $\geq \min(r, 2N_0)$, giving $\lambda_2 \geq \min(r, 2N_0)$. $\square$

Remark 9. *Theorem 27 is the direct generalization of the valuation-regime case of Theorem 24 (which has $m = 1$, $g = 1$, $N_0 = qk$). The pairwise regime ($N_0 \geq r$) is vacuous for $k, m \geq 2$: one would need $\max(pm/g, qk/g) \geq p^k + q^m$, which requires $q \cdot \max(k, m)/g \geq q^m$, i.e. $\max(k, m)/g \geq q^{m-1}$; impossible for $m \geq 2$ and $q \geq 3$. T88b verifies Theorem 27 on 40 triples ($p \leq 7$, $k, m \in \{2, \dots, 9\}$, $a, b \leq 2000$, zero failures).*

Remark 10 (Mixed witnesses in the pairwise regime). *Theorems 25 and 26 require $p_c = r$ to be the largest prime. When the group- c prime is not the largest, neither the valuation-regime formula nor the three-pure formula $\min(N_0, N_1, N_2)$ is guaranteed correct: a fully mixed (no-zero-coordinate) witness can achieve a smaller norm than any pure branch.*

The canonical example is $(5, 27, 32)$, i.e. $p_a = 5 > p_b = 3 > p_c = 2$, $k = 1$, $m = 3$, $n = 5$. The three pure branches give $N_0 = N_1 = 15$, $N_2 = 25$, so $\min(N_0, N_1, N_2) = 15$. But the mixed vector $(\varphi_{p_a}, \varphi_{p_b}, \varphi_{p_c}) = (-1, 2, 1)$ satisfies $1 \cdot (-1) + 3 \cdot 2 = 5 \cdot 1$, has $W = 3 \neq 0$, and achieves norm $\max(5, 6, 2) = 6$. Script T70 verifies $\text{nd}_{\text{val}}(5, 27) = 6$ by exhaustive search. In such cases the exact nd is computed by the LLL algorithm of Theorem 17.

Theorem 28 (Exact nd in pairwise regime via Bezout). *Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ (one prime per group), exponents k, m, n , and let $p_L = \max(p, q, r)$. Write the lattice constraint as $k_p\varphi_p + k_q\varphi_q + k_r\varphi_r = 0$ where $k_p = k$, $k_q = m$, $k_r = -n$. Let*

$N = \text{minimum norm of the } \varphi_{p_L}=0 \text{ sub-problem,}$

$$B = \min \left\{ \max_{p_i \neq p_L} p_i |\varphi_i| \mid k_{p_L} \cdot (\pm 1) + \sum_{p_i \neq p_L} k_{p_i} \varphi_i = 0, W \neq 0 \right\}$$

be the minimum weighted norm over all $\varphi_{p_L} = \pm 1$ solutions (set $B = \infty$ if none exist, i.e. $\gcd(k_{\text{other1}}, k_{\text{other2}})$ does not divide k_{p_L}).

In the pairwise regime $N > p_L$:

- (i) $\text{nd}_{\text{val}}(a, b) \geq p_L$ (lower bound, since $\varphi_{p_L} = 0$ gives norm $\geq N > p_L$ and $\varphi_{p_L} \neq 0$ gives norm $\geq p_L$);
- (ii) $\text{nd}_{\text{val}}(a, b) = p_L$ if and only if $B \leq p_L$;
- (iii) $\text{nd}_{\text{val}}(a, b) = \min(N, B)$ when $B > p_L$ (the exact value is either the pure-branch norm N or the Bezout-family minimum B , whichever is smaller).

In particular, the unified exact formula for all one-prime-per-group triples (both regimes) is

$$\text{nd}_{\text{val}}(a, b) = \min(N, \max(p_L, B)).$$

Proof. Lower bound (i): already noted above ($\varphi_{p_L} = 0$ forces norm $\geq N$; $\varphi_{p_L} \neq 0$ forces $p_L |\varphi_{p_L}| \geq p_L$).

For (ii): if $B \leq p_L$, the Bezout witness $\varphi_{p_L} = \pm 1$ with the optimal partner achieves norm $\max(p_L, B) = p_L$. Conversely, if $\text{nd}_{\text{val}} = p_L$, then a minimising vector φ has norm p_L , forcing $|\varphi_{p_L}| \leq 1$; the $\varphi_{p_L} = 0$ case gives norm $\geq N > p_L$, so $|\varphi_{p_L}| = 1$ and that vector is a Bezout witness with max other-prime norm $\leq p_L$, i.e. $B \leq p_L$.

For (iii): $\text{nd}_{\text{val}} = \min(\text{norm from } \varphi_{p_L} = 0, \text{ norm from } |\varphi_{p_L}| = 1)$ since $|\varphi_{p_L}| \geq 2$ gives norm $\geq 2p_L > \min(N, B)$. The first gives N and the second gives $\max(p_L, B) = B$ (since $B > p_L$), so $\text{nd}_{\text{val}} = \min(N, B)$. The unified formula $\min(N, \max(p_L, B))$ covers both sub-cases and the valuation regime ($N \leq p_L$, where it reduces to N). $\square$

Remark 11. *Script T72 verifies Theorem 28 on all 229 one-prime-per-group pairwise-regime triples in $[1, 600]$: the Bezout condition $B \leq p_L$ correctly predicts $\text{nd} = p_L$ in all 223 such cases, and fails in the 6 exceptions (e.g. $(5, 27)$, $(7, 25)$, $(169, 343)$) where $B > p_L$ and $\text{nd} = \min(N, B)$.*

Remark 12 (T73: high-quality triples). *Script T73 scans one-prime-per-group triples $p_a^k + p_b^m \leq 5 \times 10^5$ for quality > 1 and finds exactly 10, listed by decreasing quality:*

<i>Triple</i>	<i>quality</i>	<i>nd</i>	<i>p_L</i>	<i>nd vs p_L</i>	<i>branch</i>
(3, 125)	1.427	9	5	>	pairwise, $B = N_0 = 9$
(7, 32761)	1.327	105	181	<	$\text{val}(\varphi_{p_L} = 0)$, $N_2 = 105$
(13, 243)	1.273	13	13	=	pairwise, $B = 9 \leq 13$
(169, 343)	1.199	21	13	>	pairwise, $B = \infty$, $N = 21$
(32, 49)	1.176	7	7	=	pairwise, $B = 6 \leq 7$
(751, 130321)	1.149	323	751	<	$\text{val}(\varphi_{p_L} = 0)$, $N_1 = 323$
(139, 2048)	1.143	33	139	<	$\text{val}(\varphi_{p_L} = 0)$, $N_1 = 33$
(128, 4913)	1.095	71	71	=	pairwise, $B = 71 = p_L$
(4, 121)	1.027	10	11	<	$\text{val}(\varphi_{p_L} = 0)$, $N_2 = 10$
(5, 27)	1.019	6	5	>	pairwise, $B = 6 > p_L$

All 10 are correctly classified by Theorem 28 and Corollary 26. Four triples satisfy $\text{nd} < p_L$: the winning branch is the $\text{val}(\varphi_{p_L} = 0)$ case, which zeroes the largest prime-group coordinate (rather than the c -group as in the standard valuation-regime). Note that quality and nd are structurally independent: nd is determined by (p_a, p_b, p_c, k, m) while quality depends on n .

Corollary 29 (Minimum depends only on the smallest c -prime). *Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$ (one prime each), and P_c any finite set of primes with $r_1 = \min P_c > q > p$. Set $g = \gcd(k, m)$ where $k = v_p(a)$, $m = v_q(b)$. Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(r_1, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

In particular, the formula depends on c only through $r_1 = \min P_c$: the higher primes in P_c and their valuations are irrelevant.

Proof. Let $N_0 = \max(pm/g, qk/g)$. *Lower bound.* If all P_c -coordinates of φ are zero, the constraint reduces to the $\omega^* = 2$ problem on $\{p, q\}$, giving norm $\geq N_0$ (identical to the $\varphi_r = 0$ case in Theorem 25). If at least one coordinate φ_r with $r \in P_c$ is nonzero, then $r|\varphi_r| \geq r \geq r_1$. Hence $\text{nd}_{\text{val}}(a, b) \geq \min(r_1, N_0)$.

Upper bound. The P_c -zero witness (from Theorem 25) achieves norm N_0 . The pairwise-regime bound $\text{nd}_{\text{val}} \leq r_1$ follows from Theorem 25 applied to the sub-triple $(a, b, r_1^{v_{r_1}(c)})$ (which has $P_c = \{r_1\}$, the same lower bound, and the same N_0). $\square$

The next group of higher-dimensional type formulas has a common proof boundary: the displayed witnesses and the cases with a zero coordinate are proved, but the all-nonzero case has only finite verification. They are therefore printed as candidate formulas rather than theorems.

Candidate Formula 1 (Exact nd_{val} for $\omega^* = 4$ with two c -group primes). *Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, and $P_c = \{r, s\}$ (all four primes distinct), with $k = v_p(a)$, $m = v_q(b)$, $j = v_r(c)$, $n = v_s(c)$. Define*

$$B_x := \min_{\substack{j\varphi_r + n\varphi_s = x \\ \varphi_r, \varphi_s \in \mathbb{Z}}} \max(r|\varphi_r|, s|\varphi_s|) \quad (= \infty \text{ if } \gcd(j, n) \nmid x).$$

Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(\text{nd}_3(p^k, q^m, r^j), \quad \text{nd}_3(p^k, q^m, s^n), \quad \max(q, B_m), \quad \max(p, B_k)\right),$$

where $\text{nd}_3(\cdot)$ denotes the one-prime-per-group formula from Theorem 28.

Evidence. Upper bound. Each branch gives an explicit non-degenerate witness:

- $\varphi_s = 0$ branch: the optimal vector for the sub-triple (p^k, q^m, r^j) with $P_c = \{r\}$ has norm $\text{nd}_3(p^k, q^m, r^j)$ and satisfies $k\varphi_p + m\varphi_q = j\varphi_r$ with $W = \varphi_q - \varphi_p \neq 0$.
- $\varphi_r = 0$ branch: analogously for (p^k, q^m, s^n) .
- $\varphi_p = 0, \varphi_q = \pm 1$ branch: set $j\varphi_r + n\varphi_s = \pm m$ with the pair (φ_r, φ_s) achieving B_m ; then $\varphi_p = 0$ satisfies the constraint, $W = \varphi_q \neq 0$, and the norm is $\max(q, B_m)$.
- $\varphi_q = 0, \varphi_p = \pm 1$ branch: analogously, achieving $\max(p, B_k)$.

Lower bound. If $\varphi_s = 0$: norm $\geq \text{nd}_3(p^k, q^m, r^j)$ by Theorem 28 applied to the three-prime sub-problem. If $\varphi_r = 0$: analogously, norm $\geq \text{nd}_3(p^k, q^m, s^n)$ by Theorem 28. If $\varphi_p = 0$ and $\varphi_q \neq 0$: norm $\geq q|\varphi_q| \geq q$, and the pair (φ_r, φ_s) satisfies $j\varphi_r + n\varphi_s = m\varphi_q$, giving norm $\geq \max(q, B_{m|\varphi_q|}) \geq \max(q, B_m)$. If $\varphi_q = 0$ and $\varphi_p \neq 0$: analogously $\geq \max(p, B_k)$. The case where all four coordinates are nonzero is verified computationally (script T74, 431 cases in $[1, 120)$, 0 failures): no such witness achieves a smaller norm than the minimum of the four branches. $\square$

Remark 13. Script T74 verifies Candidate Formula 1 on all 431 such triples in $[1, 120)$ (initial run: 262 triples in $[1, 80)$): 0 failures in both ranges. The $\varphi_q = 0$ branch is the minimum in 25 cases (e.g. $(37, 71)$: $p = 37, q = 71, c = 2^2 \cdot 3^3, \text{nd}_{\text{val}} = 37 = \max(37, B_1 = 3)$); the $\varphi_p = 0$ branch wins in 5 cases. The two $\varphi_{c\text{-prime}} = 0$ branches suffice in 223/262 cases.

Candidate Formula 2 (Exact nd_{val} for type $(2, 1, 1)$ triples). Let $a + b = c$ be coprime with $P_a = \{p, q\}$ ($p < q$, exponents $k_1 = v_p(a), k_2 = v_q(a)$), $P_b = \{r\}$ (exponent m), $P_c = \{s\}$ (exponent n), all four primes distinct. Define the P_a – P_b interaction norm

$$N_{s=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k_1\varphi_p + k_2\varphi_q + m\varphi_r = 0, \varphi_r \neq \varphi_p + \varphi_q\}$$

(zeroing the P_c -coordinate; subsumes the within-group norm and the two P_a – P_b pairwise norms), and the P_a – P_c interaction norm

$$N_{r=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, s|\varphi_s|) : k_1\varphi_p + k_2\varphi_q = n\varphi_s, \varphi_s \neq 0, \varphi_p + \varphi_q \neq 0\}$$

(zeroing P_b ; subsumes the two P_a – P_c pairwise norms). Let $\text{nd}_3(p^{k_1}, r^m, s^n)$ and $\text{nd}_3(q^{k_2}, r^m, s^n)$ denote the exact nd_{val} values for the $\omega^* = 3$ sub-problems at primes (p, r, s) and (q, r, s) with exponent vectors (k_1, m, n) and (k_2, m, n) (computed via Theorem 28; each subsumes the corresponding pairwise norms and the P_b – P_c pairwise norm). Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(N_{s=0}, \quad N_{r=0}, \quad \text{nd}_3(p^{k_1}, r^m, s^n), \quad \text{nd}_3(q^{k_2}, r^m, s^n)\right).$$

Evidence. Upper bound. Each branch provides a constructive non-degenerate witness: $N_{s=0}$ and $N_{r=0}$ use a Bezout construction on the P_a -sub-lattice (analogous to Candidate Formula 1); the $\omega^* = 3$ branches use Theorem 28 directly.

Lower bound. Any non-degenerate $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s)$ with $k_1\varphi_p + k_2\varphi_q + m\varphi_r = n\varphi_s$ and $W = \varphi_r - (\varphi_p + \varphi_q) \neq 0$ satisfies at least one of:

- $\varphi_s = 0$: norm $\geq N_{s=0}$ by definition.
- $\varphi_r = 0, \varphi_s \neq 0$: norm $\geq N_{r=0}$ by definition.
- $\varphi_q = 0$: φ lies in the $\omega^* = 3$ sub-lattice for (p^{k_1}, r^m, s^n) , so norm $\geq \text{nd}_3(p^{k_1}, r^m, s^n)$.
- $\varphi_p = 0$: analogously norm $\geq \text{nd}_3(q^{k_2}, r^m, s^n)$.
- All four coordinates non-zero: verified computationally (script T77, 672 triples in $[4, 150) \times [1, 150)$, 0 failures).

The first four cases cover all vectors with at least one zero coordinate; note that a vector may fall in more than one case, but the minimum suffices. $\square$

Remark 14. *The four branches correspond to zeroing one of the four prime-group slots. $N_{s=0}$ subsumes the within-group norm, all P_a - P_b pairwise norms, and any vector with $\varphi_r = \varphi_s = 0$. $N_{r=0}$ subsumes all P_a - P_c pairwise norms. Each $\omega^* = 3$ branch subsumes the P_b - P_c pairwise norm and the remaining two pairwise norms for its P_a -prime. Script T77 (672 triples, 0 failures) confirms that no all-nonzero vector can improve on this minimum: $N_{s=0}$ is the minimum in 582 cases (87%), $\text{nd}_3(p^{k_1}, r^m, s^n)$ in 378, $N_{r=0}$ in 70, $\text{nd}_3(q^{k_2}, r^m, s^n)$ in 172.*

Candidate Formula 3 (Exact nd_{val} for type $(1, 2, 1)$ triples). *Let $a + b = c$ be coprime with $P_a = \{p\}$ (exponent k), $P_b = \{q, r\}$ ($q < r$, exponents $j = v_q(b)$, $m = v_r(b)$), $P_c = \{s\}$ (exponent n), all four primes distinct. Define the P_a - P_b interaction norm*

$$N_{s=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k\varphi_p + j\varphi_q + m\varphi_r = 0, (\varphi_q + \varphi_r) \neq \varphi_p\}$$

(zeroing the P_c -coordinate; subsumes the within-group P_b norm and the two P_a - P_b pairwise norms), and the P_b - P_c interaction norm

$$N_{p=0} := \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : j\varphi_q + m\varphi_r = n\varphi_s, \varphi_q + \varphi_r \neq 0\}$$

(zeroing P_a ; subsumes the two P_b - P_c pairwise norms). Let $\text{nd}_3(p^k, q^j, s^n)$ and $\text{nd}_3(p^k, r^m, s^n)$ denote the $\omega^ = 3$ sub-problems (Theorem 28; each subsumes the P_a - P_c pairwise norm and the corresponding P_b - P_c norm). Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(N_{s=0}, N_{p=0}, \text{nd}_3(p^k, q^j, s^n), \text{nd}_3(p^k, r^m, s^n)\right).$$

Evidence. The Wronskian is $W = (\varphi_q + \varphi_r) - \varphi_p$ and the constraint is $k\varphi_p + j\varphi_q + m\varphi_r = n\varphi_s$. This has the same algebraic structure as Candidate Formula 2 with P_a and P_b exchanged: relabelling $p \leftrightarrow r$, $k_1 \leftrightarrow m$, $k_2 \leftrightarrow j$, and identifying the single-prime group as $P_a = \{p\}$ maps the four-branch formula of Candidate Formula 2 verbatim to the formula above. The subsumption claims are verified by setting one or two coordinates to zero in the respective minimisation problems. $\square$

Remark 15. Script T78_formula verifies Candidate Formula 3 on all 1043 type (1, 2, 1) triples with $a, b \leq 200$: 0 failures. Branch win counts (ties included): $N_{s=0}$ in 935 cases (90%), $\text{nd}_3(p^k, q^j, s^n)$ in 567, $N_{p=0}$ in 395, $\text{nd}_3(p^k, r^m, s^n)$ in 265. One spot-check case ($a = 2^7$, $b = 3 \cdot 23$, $c = 197$) requires $|\varphi_3| = 7$ in the optimal witness via $N_{s=0}$ ($\text{nd}_{\text{val}} = 21$).

The same recursive strategy extends to $\omega = 5$.

Candidate Formula 4 (Exact nd_{val} for type (2, 2, 1) triples). Let $a + b = c$ be coprime with $P_a = \{p, q\}$ (exponents k_1, k_2), $P_b = \{r, s\}$ (exponents j_1, j_2), $P_c = \{u\}$ (exponent n), five distinct primes. Define the P_a – P_b interaction norm

$$N_{u=0} := \min \left\{ \max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : \begin{array}{l} k_1\varphi_p + k_2\varphi_q + j_1\varphi_r + j_2\varphi_s = 0, \\ (\varphi_r + \varphi_s) \neq (\varphi_p + \varphi_q) \end{array} \right\}.$$

Let $B_{\varphi_q=0}$ (resp. $B_{\varphi_p=0}$) be the exact nd_{val} of the type-(1, 2, 1) sub-problem on primes (p, r, s, u) with exponents (k_1, j_1, j_2, n) (resp. (q, r, s, u) with (k_2, j_1, j_2, n)), computed via Candidate Formula 3. Let $B_{\varphi_s=0}$ (resp. $B_{\varphi_r=0}$) be the exact nd_{val} of the type-(2, 1, 1) sub-problem on primes (p, q, r, u) with exponents (k_1, k_2, j_1, n) (resp. (p, q, s, u) with (k_1, k_2, j_2, n)), computed via Candidate Formula 2. Then

$$\text{nd}_{\text{val}}(a, b) = \min(N_{u=0}, B_{\varphi_q=0}, B_{\varphi_p=0}, B_{\varphi_s=0}, B_{\varphi_r=0}).$$

Evidence. Upper bound. Each branch supplies a constructive non-degenerate witness: $N_{u=0}$ by Bezout search on the four-prime homogeneous lattice; $B_{\varphi_q=0}$, $B_{\varphi_p=0}$ by Candidate Formula 3; $B_{\varphi_s=0}$, $B_{\varphi_r=0}$ by Candidate Formula 2.

Lower bound. Any non-degenerate $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s, \varphi_u)$ with $k_1\varphi_p + k_2\varphi_q + j_1\varphi_r + j_2\varphi_s = n\varphi_u$ and $W = (\varphi_r + \varphi_s) - (\varphi_p + \varphi_q) \neq 0$ satisfies at least one of:

- $\varphi_u = 0$: norm $\geq N_{u=0}$ by definition.
- $\varphi_q = 0$: φ lies in the type-(1, 2, 1) sub-lattice for $(p^{k_1}, r^{j_1}, s^{j_2}, u^n)$, so norm $\geq B_{\varphi_q=0}$.
- $\varphi_p = 0$: analogously norm $\geq B_{\varphi_p=0}$.
- $\varphi_s = 0$: φ lies in the type-(2, 1, 1) sub-lattice for $(p^{k_1}, q^{k_2}, r^{j_1}, u^n)$, so norm $\geq B_{\varphi_s=0}$.
- $\varphi_r = 0$: analogously norm $\geq B_{\varphi_r=0}$.
- All five coordinates non-zero: verified computationally (script T79, 1425 triples in $[4, 300) \times [4, 300)$, spot-check 60, 0 failures).

□

Remark 16. The five branches correspond to zeroing each prime-group slot in turn. $N_{u=0}$ subsumes the within- P_a , within- P_b , and all P_a – P_b cross-group pairwise norms; the type-(1, 2, 1) and type-(2, 1, 1) sub-problem branches subsume all P_a – P_c and P_b – P_c interactions. Script T79 (1425 triples, 0 failures, spot-check against brute-force): $N_{u=0}$ achieves the minimum in 1405 cases (98.6%); $B_{\varphi_s=0}$ or $B_{\varphi_r=0}$ wins in the remaining 20 cases.

Candidate Formula 5 (Exact nd_{val} for type $(1, 2, 2)$ triples). Let $a + b = c$ be coprime with $P_a = \{p\}$ (exponent k), $P_b = \{q, r\}$ (exponents j, m), $P_c = \{s, t\}$ (exponents j_1, j_2), five distinct primes. The Wronskian is $W = (\varphi_q + \varphi_r) - \varphi_p$. Define the P_b – P_c interaction norm

$$N_{p=0} := \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : j\varphi_q + m\varphi_r - j_1\varphi_s - j_2\varphi_t = 0, \varphi_q + \varphi_r \neq 0\}.$$

Let $B_{\varphi_r=0}$ (resp. $B_{\varphi_q=0}$) be the exact nd_{val} of the type- $(1, 1, 2)$ sub-problem on primes (p, q, s, t) with exponents (k, j, j_1, j_2) (resp. (p, r, s, t) with (k, m, j_1, j_2)), via Candidate Formula 1. Let $B_{\varphi_t=0}$ (resp. $B_{\varphi_s=0}$) be the exact nd_{val} of the type- $(1, 2, 1)$ sub-problem on primes (p, q, r, s) with exponents (k, j, m, j_1) (resp. (p, q, r, t) with (k, j, m, j_2)), via Candidate Formula 3. Then

$$\text{nd}_{\text{val}}(a, b) = \min(N_{p=0}, B_{\varphi_r=0}, B_{\varphi_q=0}, B_{\varphi_t=0}, B_{\varphi_s=0}).$$

Evidence. Upper bound. Each branch supplies a constructive witness: $N_{p=0}$ by Bezout search on the four-prime homogeneous lattice; $B_{\varphi_r=0}, B_{\varphi_q=0}$ by Candidate Formula 1; $B_{\varphi_t=0}, B_{\varphi_s=0}$ by Candidate Formula 3.

Lower bound. Any non-degenerate $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s, \varphi_t)$ with $k\varphi_p + j\varphi_q + m\varphi_r = j_1\varphi_s + j_2\varphi_t$ and $W = (\varphi_q + \varphi_r) - \varphi_p \neq 0$ satisfies at least one of:

- $\varphi_p = 0$: norm $\geq N_{p=0}$ by definition.
- $\varphi_r = 0$: φ lies in the type- $(1, 1, 2)$ sub-lattice for $(p^k, q^j, s^{j_1}, t^{j_2})$, so norm $\geq B_{\varphi_r=0}$.
- $\varphi_q = 0$: analogously norm $\geq B_{\varphi_q=0}$.
- $\varphi_t = 0$: φ lies in the type- $(1, 2, 1)$ sub-lattice for $(p^k, \{q^j, r^m\}, s^{j_1})$, so norm $\geq B_{\varphi_t=0}$.
- $\varphi_s = 0$: analogously norm $\geq B_{\varphi_s=0}$.
- All five coordinates non-zero: verified computationally (script T80, 996 triples in $[2, 150) \times [4, 150)$, spot-check 50, 0 failures).

□

Remark 17. Script T80 (996 triples, 0 failures): $N_{p=0}$ achieves the minimum in 869 cases (87.2%); $B_{\varphi_r=0}$ wins in 72 cases (7.2%), $B_{\varphi_q=0}$ in 52 (5.2%), $B_{\varphi_t=0}$ in 3 (0.3%).

Candidate Formula 6 (Exact nd_{val} for type $(2, 1, 2)$ triples). Let $a + b = c$ be coprime with $P_a = \{p, q\}$ (exponents k_1, k_2), $P_b = \{r\}$ (exponent m), $P_c = \{s, t\}$ (exponents j_1, j_2). By relabelling $a \leftrightarrow b$ (so $P_a \leftrightarrow P_b$), this is equivalent to a type- $(1, 2, 2)$ triple with the same five primes. Applying Candidate Formula 5 with the label swap gives

$$\text{nd}_{\text{val}}(a, b) = \min(N_{r=0}, B_{\varphi_q=0}, B_{\varphi_p=0}, B_{\varphi_t=0}, B_{\varphi_s=0}),$$

where $N_{r=0}$ is the P_a – P_c homogeneous interaction (zeroing φ_r), $B_{\varphi_q=0}$ and $B_{\varphi_p=0}$ are type- $(1, 1, 2)$ sub-problems via Candidate Formula 1, and $B_{\varphi_t=0}, B_{\varphi_s=0}$ are type- $(2, 1, 1)$ sub-problems via Candidate Formula 2.

Candidate Formula 7 (Exact nd_{val} for type $(1, 1, 3)$ triples). *Let $a + b = c$ be coprime with $P_a = \{p\}$ (exponent k), $P_b = \{q\}$ (exponent m), $P_c = \{r, s, t\}$ (exponents n_1, n_2, n_3), five distinct primes. The constraint is $k\varphi_p + m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$ and $W = \varphi_q - \varphi_p$. Define the one-sided interaction norms*

$$\begin{aligned} N_{p=0} &:= \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t, \varphi_q \neq 0\}, \\ N_{q=0} &:= \min\{\max(p|\varphi_p|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : k\varphi_p = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t, \varphi_p \neq 0\}. \end{aligned}$$

Let $\text{nd}_4(\cdot)$ denote the type- $(1, 1, 2)$ formula (Candidate Formula 1). Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(\text{nd}_4(p^k, q^m, s^{n_2}, t^{n_3}), \text{nd}_4(p^k, q^m, r^{n_1}, t^{n_3}), \text{nd}_4(p^k, q^m, r^{n_1}, s^{n_2}), N_{p=0}, N_{q=0}\right).$$

Evidence. Upper bound. The three nd_4 branches use Candidate Formula 1 on the sub-triple obtained by setting one c -prime coordinate to zero; $N_{p=0}$ takes $\varphi_p = 0$ with $W = \varphi_q \neq 0$; $N_{q=0}$ takes $\varphi_q = 0$ with $W = -\varphi_p \neq 0$.

Lower bound. Any non-degenerate φ satisfies at least one of:

- $\varphi_r = 0$: φ lies in the type- $(1, 1, 2)$ sub-lattice for $(p^k, q^m, s^{n_2}t^{n_3})$, so $\text{norm} \geq \text{nd}_4(p^k, q^m, s^{n_2}, t^{n_3})$.
- $\varphi_s = 0$: analogously $\text{norm} \geq \text{nd}_4(p^k, q^m, r^{n_1}, t^{n_3})$.
- $\varphi_t = 0$: analogously $\text{norm} \geq \text{nd}_4(p^k, q^m, r^{n_1}, s^{n_2})$.
- $\varphi_p = 0$: $W = \varphi_q \neq 0$, so $m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$ with $\varphi_q \neq 0$, hence $\text{norm} \geq N_{p=0}$.
- $\varphi_q = 0$: $W = -\varphi_p \neq 0$, so $k\varphi_p = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$ with $\varphi_p \neq 0$, hence $\text{norm} \geq N_{q=0}$.
- All five coordinates non-zero: verified computationally (script T81, 729 triples with $a, b \leq 300$, spot-check 60, 0 failures).

□

Remark 18. *Script T81 verifies Candidate Formula 7 on all 729 type- $(1, 1, 3)$ triples with $a, b \leq 300$, spot-check 60: 0 failures. Branch win counts: $\text{nd}_4(\cdot, r^{n_1}, s^{n_2})$ (zeroing φ_t) in 669 cases (91.8%); $\text{nd}_4(\cdot, r^{n_1}, t^{n_3})$ in 32 (4.4%); $N_{q=0}$ in 16 (2.2%); $N_{p=0}$ in 12 (1.6%); $\text{nd}_4(\cdot, s^{n_2}, t^{n_3})$ in 0 cases.*

Candidate Formula 8 (Exact nd_{val} for type $(1, 3, 1)$ triples). *Let $a + b = c$ be coprime with $P_a = \{p\}$ (exponent k), $P_b = \{q, r, s\}$ (exponents n_1, n_2, n_3), $P_c = \{t\}$ (exponent m), five distinct primes. The Wronskian is $W = (\varphi_q + \varphi_r + \varphi_s) - \varphi_p$. Define the P_a - P_b homogeneous norm*

$$N_{t=0} := \min\left\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : \begin{aligned} &k\varphi_p + n_1\varphi_q + n_2\varphi_r + n_3\varphi_s = 0, \\ &\varphi_q + \varphi_r + \varphi_s \neq \varphi_p \end{aligned} \right\}$$

and the P_b - P_c interaction norm

$$N_{p=0} := \min \left\{ \max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : \begin{array}{l} n_1\varphi_q + n_2\varphi_r + n_3\varphi_s = m\varphi_t, \\ \varphi_q + \varphi_r + \varphi_s \neq 0 \end{array} \right\}.$$

Let $B_{\varphi_q=0}$ (resp. $B_{\varphi_r=0}$, $B_{\varphi_s=0}$) be the exact nd_{val} of the type-(1, 2, 1) sub-problem on primes (p, r, s, t) with exponents (k, n_2, n_3, m) (resp. (p, q, s, t) with (k, n_1, n_3, m) ; (p, q, r, t) with (k, n_1, n_2, m)), computed via Candidate Formula 3. Then

$$\text{nd}_{\text{val}}(a, b) = \min(N_{t=0}, N_{p=0}, B_{\varphi_q=0}, B_{\varphi_r=0}, B_{\varphi_s=0}).$$

Evidence. Upper bound. $N_{t=0}$ takes $\varphi_t = 0$ with $W \neq 0$; $N_{p=0}$ takes $\varphi_p = 0$ with $W = \varphi_q + \varphi_r + \varphi_s \neq 0$; each $B_{\varphi_\bullet=0}$ uses Candidate Formula 3 on the sub-lattice obtained by setting one b -prime coordinate to zero.

Lower bound. Any non-degenerate φ satisfies at least one of:

- $\varphi_t = 0$: norm $\geq N_{t=0}$ by definition.
- $\varphi_p = 0$: $W = \varphi_q + \varphi_r + \varphi_s \neq 0$, so norm $\geq N_{p=0}$.
- $\varphi_q = 0$: φ lies in the type-(1, 2, 1) sub-lattice for $(p^k, r^{n_2}, s^{n_3}, t^m)$, so norm $\geq B_{\varphi_q=0}$.
- $\varphi_r = 0$: analogously norm $\geq B_{\varphi_r=0}$.
- $\varphi_s = 0$: analogously norm $\geq B_{\varphi_s=0}$.
- All five coordinates non-zero: verified computationally (script T82, 843 triples with $a, b \leq 300$, spot-check 60, 0 failures).

□

Remark 19. Script T82 verifies Candidate Formula 8 on all 843 type-(1, 3, 1) triples with $a, b \leq 300$, spot-check 60: 0 failures. Branch win counts: $N_{t=0}$ in 803 cases (95.3%); $N_{p=0}$ in 40 (4.7%); the three $B_{\varphi_\bullet=0}$ branches in 0 cases. The empirical two-branch formula $\min(N_{t=0}, N_{p=0})$ holds on all 843 surveyed triples; independently confirmed by a direct spot-check (seed 91, 60 triples, 0 failures).

Candidate Formula 9 (Exact nd_{val} for type (3, 1, 1) triples). Let $a + b = c$ be coprime with $P_a = \{p, q, r\}$ (exponents k_1, k_2, k_3), $P_b = \{s\}$ (exponent m), $P_c = \{t\}$ (exponent n), five distinct primes. The exact nd_{val} formula is obtained from Candidate Formula 8 by the $P_a \leftrightarrow P_b$ label swap:

$$\text{nd}_{\text{val}}(a, b) = \min(\tilde{N}_{t=0}, \tilde{N}_{s=0}, \tilde{B}_{\varphi_p=0}, \tilde{B}_{\varphi_q=0}, \tilde{B}_{\varphi_r=0}),$$

where $\tilde{N}_{t=0}$ is the P_a - P_b homogeneous norm (zero φ_t , $W \neq 0$), $\tilde{N}_{s=0}$ is the P_a - P_c interaction norm (zero φ_s , $\varphi_p + \varphi_q + \varphi_r \neq 0$), and $\tilde{B}_{\varphi_p=0}$, $\tilde{B}_{\varphi_q=0}$, $\tilde{B}_{\varphi_r=0}$ are type-(1, 2, 1) sub-problems via Candidate Formula 3.

Evidence. The constraint for type (3, 1, 1) is $k_1\varphi_p + k_2\varphi_q + k_3\varphi_r + m\varphi_s = n\varphi_t$ with $W = \varphi_s - (\varphi_p + \varphi_q + \varphi_r)$. Under $a \leftrightarrow b$ this maps to the type-(1, 3, 1) setup of Candidate Formula 8 with $P_a \leftrightarrow P_b$ and $W \rightarrow -W$; non-degeneracy is symmetric in sign of W , so Candidate Formula 8 applies verbatim. □

Zero-coordinate minimizer property

The results in this and the following subsection that are printed as *Candidate Formula* are finite-verification formulas: their displayed witnesses and branch reductions are proved, but the all-nonzero coordinate case has only been checked on the stated finite sample. They must not be used as unconditional theorems.

Theorem 30 (Squarefree minimizer support). *Let $a + b = c$ be a squarefree coprime triple ($v_p(abc) = 1$ for all p) with $\omega^* \geq 3$ distinct primes. Let $m_1 \leq m_2 \leq m_3$ be the sorted group minima $\min(P_a), \min(P_b), \min(P_c)$. Then:*

- (i) $\text{nd}(a, b) \leq m_2$ (Corollary 16: cross-group pair (m_1, m_2) with $\gcd = 1$ gives a non-degenerate vector of norm $\max(m_1, m_2) = m_2$).
- (ii) Every nd-minimizer φ satisfies $\varphi_p = 0$ for every prime $p > m_2$. (Proof: $p|\varphi_p| \leq \text{nd} \leq m_2 < p$, so $|\varphi_p| < 1$, forcing $\varphi_p = 0$.)
- (iii) At least $\omega^* - 2$ coordinates of every minimizer are zero. In particular $\text{nd}(a, b) = \min_p B_p$ (zero-coordinate property, proved for all squarefree triples of any $\omega^* \geq 3$).

Proof. (i) is the squarefree case of Corollary 16 (all valuations 1, $\gcd = 1$, $\text{norm} = m_2$).

(ii) For any prime $p > m_2$: since $m_1 \leq m_2 < m_3 \leq p$ and $\text{nd} \leq m_2$, we have $p|\varphi_p| \leq \text{nd} \leq m_2 < p$, giving $|\varphi_p| < 1$, hence $\varphi_p = 0$.

(iii) Among the ω^* primes, at most two satisfy $p \leq m_2$ (namely m_1 and m_2 themselves; all other group elements exceed m_2 by minimality of group minima within their groups). Hence at least $\omega^* - 2$ primes exceed m_2 and are forced to zero. For $\omega^* \geq 3$: $\omega^* - 2 \geq 1$, so every minimizer has at least one zero coordinate, i.e. $\text{nd}(a, b) \geq \min_p B_p$. The reverse inequality $\text{nd} \leq \min_p B_p$ is immediate (each B_p is a minimum over a sub-lattice). $\square$

Remark 20. *The squarefree hypothesis is essential. For non-squarefree $\omega^* = 3$ triples, nd can exceed all primes in the triple (e.g. $(5, 27, 32)$: primes $\{2, 3, 5\}$, $\text{nd} = 6 > 5$), so the integrality argument fails and all-nonzero minimizers can occur.*

The individual exact nd formulas (Candidate Formulas 2–9) share a common structural feature: in every verified case, the minimum non-degenerate norm is achieved by a lattice vector with at least one prime coordinate equal to zero. We record the non-squarefree instances as a finite-verification candidate.

Candidate Formula 10 (Zero-coordinate minimizer property). *Let $a + b = c$ be a coprime triple with ω^* distinct primes. Define, for each prime $p \in P_a \cup P_b \cup P_c$,*

$$B_p := \min_q \{ \max_q q|\varphi_q| : \varphi \in F_{\text{val}}(a, b), \varphi_p = 0, W(\varphi) \neq 0 \}.$$

In parts (ii)–(iv), $F(a, b)$ is the valuation-coordinate lattice $F_{\text{val}}(a, b)$ and nd means nd_{val} . Then:

- (i) For squarefree triples of any $\omega^* \geq 3$: $\text{nd}(a, b) = \min_p B_p$ (proved, Theorem 30).
- (ii) For non-squarefree $\omega^* = 4$: $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ (supported by Candidate Formulas 2, 3, and 1, each with the all-nonzero case verified only on the stated finite sample).

(iii) For non-squarefree $\omega^* \in \{5, 6\}$: $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ on the finite samples described below. Both cases use the p_k -forcing argument (primes labelled $p_1 < \dots < p_k$, $k = \omega^*$): any all-nonzero φ with $\|\varphi\| < B_{p_k}$ satisfies $p_k |\varphi_{p_k}| \leq \|\varphi\| < p_k$, forcing $\varphi_{p_k} = 0$. Thus it suffices to verify $B_{p_k} \leq p_k$.

$\omega^* = 5$ (2446 non-squarefree triples, $a, b \leq 150$): 2019 (82.5%) via a squarefree cross-group pair in $\{p_1, \dots, p_4\}$ (analytic, T86bis sub-case B1); 427 (17.5%) via $B_{p_5} \leq p_5$ directly (T86bis, BB=8, 0 violations, 0 skips).

$\omega^* = 6$ (1959 non-squarefree triples, $a, b \leq 200$): 1920 (98.0%) via a squarefree cross-group pair in $\{p_1, \dots, p_5\}$ (analytic, T85bis sub-case B1); 39 (2.0%) via $B_{p_6} \leq p_6$ directly (T85bis, BB=8, 0 violations, 0 skips).

(iv) For non-squarefree $\omega^* = 3$: the identity fails. The triple (5, 27, 32) has $\text{nd}_{\text{val}} = 6$ (achieved by $\varphi = (-1, 2, 1)$, all coordinates non-zero), while $\min_p B_p = 15$.

Evidence. Part (i). Theorem 30: integrality forces $\varphi_p = 0$ for all $p > \text{nd} \leq m_2$ (second group minimum).

Part (ii). For type (2, 1, 1): Candidate Formula 2 decomposes nd into four branches each zeroing one prime ($N_{s=0}$, $N_{r=0}$, two nd -111 sub-problems with $\varphi_q = 0$ or $\varphi_p = 0$). Types (1, 2, 1) and (1, 1, 2) follow analogously from Candidate Formulas 3 and 1.

Part (iii). Label the primes $p_1 < \dots < p_k$ ($k = \omega^* \in \{5, 6\}$). The p_k -forcing argument applies uniformly: any all-nonzero $\varphi \in F_{\text{val}}(a, b)$ with $W(\varphi) \neq 0$ satisfies $p_k |\varphi_{p_k}| \leq \|\varphi\|$; if $\|\varphi\| < B_{p_k}$ then $p_k |\varphi_{p_k}| < p_k$, forcing $\varphi_{p_k} = 0$ — contradiction. Hence $\text{nd}(a, b) \geq B_{p_k}$, and equality $\text{nd}(a, b) = \min_p B_p$ holds as soon as $B_{p_k} \leq p_k$.

Sub-case B1 (analytic, 82.5% for $\omega^* = 5$; 98.0% for $\omega^* = 6$): there exist $p_i, p_j \in \{p_1, \dots, p_{k-1}\}$ with $v_{p_i}(abc) = v_{p_j}(abc) = 1$ lying in different groups. The cross-group Bézout vector (φ_{p_i} and φ_{p_j} the unique non-zero coordinates, all others 0) has $\varphi_{p_k} = 0$, $W \neq 0$, and norm $\max(p_i, p_j) \leq p_{k-1} < p_k$. Hence $B_{p_k} \leq p_{k-1} < p_k$.

Sub-case B2 ($\omega^* = 5$: 427 triples; $\omega^* = 6$: 39 triples): no squarefree cross-group pair exists in $\{p_1, \dots, p_{k-1}\}$. Brute-force search on the zero- p_k sub-lattice with $\|\varphi\|_\infty \leq 8$ verifies $B_{p_k} \leq p_k$ for all such triples (T86bis $\omega^* = 5$; T85bis $\omega^* = 6$; 0 violations, 0 skips).

Part (iv). For $(a, b, c) = (5, 27, 32)$: $P_a = \{5\}$, $P_b = \{3\}$, $P_c = \{2\}$, exponents 1, 3, 5. The vector $\varphi = (-1, 2, 1)$ satisfies the lattice constraint ($1 \cdot (-1) + 3 \cdot 2 = 5 \cdot 1$), $W = 2 - (-1) = 3 \neq 0$, and $\|\varphi\| = \max(5, 6, 2) = 6$. The three zero-coordinate branches give norms 15, 25, 15 (computed from the one-prime-per-group base case), so $\min_p B_p = 15 > 6$. $\square$

Remark 21. The failure at $\omega^* = 3$ is explained by the pairwise-Bézout regime (Theorem 28): when p_L is the largest prime and $\gcd(\text{other exponents}) \mid k_{p_L}$, the minimizer has $|\varphi_{p_L}| = 1$ and all other coordinates nonzero. For $\omega^* \geq 4$ this all-nonzero configuration at the base level is subsumed: the zero- φ_{p_L} branch reduces to a $(\omega^* - 1)$ -prime sub-problem whose own Bézout regime provides a vector at least as good.

General recursive nd algorithm

Candidate Formula 10 and the individual exact- nd results suggest the following recursion for $\text{nd}(a, b)$; its non-squarefree branches concern nd_{val} and are finite-verification claims outside the squarefree case.

Candidate Formula 11 (General recursive nd formula). *Let $a + b = c$ be a coprime triple with $\omega^* \geq 3$ distinct primes partitioned as $P_a \cup P_b \cup P_c$. For each prime $p \in P_a \cup P_b \cup P_c$, define the p -branch B_p recursively:*

- (Multi-prime reduction) *If $|P_x| \geq 2$ for the group P_x containing p : $B_p = \text{nd}(a', b')$ where (a', b', c') is the $(\omega^* - 1)$ -prime sub-problem obtained by removing prime p from P_x (its coordinate is forced to zero).*
- (Singleton interaction) *If $|P_x| = 1$: B_p is the group-interaction minimum $\min\{\max_q q|\varphi_q| : \varphi \in F_{\text{val}}(a, b), \varphi_p = 0, W(\varphi) \neq 0\}$, a direct Bezout-type minimization over the remaining two-group constraint.*

The base case ($\omega^* = 3$) is Theorem 28. Then:

- (i) *For non-squarefree $\omega^* = 4$: $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ (supported by Candidate Formulas 2, 3, 1).*
- (ii) *For non-squarefree $\omega^* \in \{5, 6\}$: $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ (empirically verified for non-squarefree; $\omega^* = 5$ both partition types $(2, 2, 1)$ and $(3, 1, 1)$, $a, b \leq 150$, T86bis; $\omega^* = 6$ all three unordered partition classes $\{4, 1, 1\}$, $\{3, 2, 1\}$, $\{2, 2, 2\}$: 1959 non-squarefree triples $a, b \leq 200$, 0 violations, T85bis; squarefree $\omega^* \in \{5, 6\}$: proved, Theorem 30).*

Evidence. Upper bound ($\text{nd} \leq \min_p B_p$). Each B_p is a minimum over a sub-lattice of $F(a, b)$, hence $\text{nd}(a, b) \leq B_p$ for every p .

Lower bound for non-squarefree $\omega^* = 4$ ($\text{nd}_{\text{val}} \geq \min_p B_p$). This is supported by Candidate Formula 10(ii): in every surveyed minimizer some coordinate is zero, placing it in one of the $\omega^* = 4$ branches. The exact-nd sub-problem formulas (Candidate Formulas 1–2) confirm the identity.

Part (ii). Computational verification; see Candidate Formula 10(ii) for the zero-coordinate property and the individual T-scripts for the exact values. $\square$

For $\omega = 3$ type $(1, 1, 1)$ we can characterize not just the minimum but also the *second* minimum non-degenerate norm.

Theorem 31 (F32a: second minimum for type $(1, 1, 1)$). *For a type $(1, 1, 1)$ triple with primes $p < q < r = p + q$, the second minimum non-degenerate norm is $r = q + 2$ (achieved by the vector $\psi = (0, q, r)$, i.e. $\varphi = (0, 1, 1)$, which has $W = pq \neq 0$ and $\|\psi\|_\infty = r$). Moreover there is no non-degenerate vector with norm strictly between q and r .*

Proof. Upper bound. The vector $\varphi = (0, 1, 1)$ satisfies the lattice constraint $S_a + S_b = S_c$ (i.e. $0 + 1 = 1$) and is non-degenerate since $S_b = 1 \neq 0 = S_a$. Its norm is $\max(0, q, r) = r = q + 2$.

Lower bound (no gap). Suppose ψ is non-degenerate with $\|\psi\|_\infty \in (q, r)$. Since $r|\varphi_r|$ must be strictly less than r , we have $\varphi_r = 0$. The constraint then gives $\varphi_q = -\varphi_p$. Non-degeneracy ($S_b \neq S_a$) forces $\varphi_p \neq 0$. But the norm is $\max(p|\varphi_p|, q|\varphi_p|, 0) = q|\varphi_p|$. For the norm to lie strictly in $(q, r) = (q, q + 2)$ we need $q < q|\varphi_p| < q + 2$, i.e. $1 < |\varphi_p| < 1 + 2/q$. Since $q \geq 3$ (the smallest odd prime gap gives $q = 3$), $2/q \leq 2/3 < 1$, so there is no integer in $(1, 1 + 2/q)$. Contradiction.

The integer core ($q + 2 \leq kq$ for all $k \geq 2$ and $q \geq 3$, with no integer k satisfying $q < kq < q + 2$) is formalized as `pasten_F32a_gap` and `pasten_F32a_upper` in the Lean kernel. $\square$

Theorem 32 (F32b: complete norm spectrum for type $(1, 1, 1)$, $q \geq 5$). *Let $(2, q, r)$ be a twin-prime triple with $q \geq 5$. The set of achievable non-degenerate norms is exactly*

$$\{kq : k \geq 1\} \cup \{kr : k \geq 1\}.$$

Proof. Achievability. The vector $\varphi = (k, -k, 0)$ has norm kq (non-degenerate since $S_b = -k \neq k = S_a$); $\varphi = (0, k, k)$ has norm kr .

Completeness. Let $(\alpha, \beta) \in \mathbb{Z}^2$ with $\alpha \neq \beta$, $\gamma := \alpha + \beta$. If $\gamma = 0$ then norm $= q|\alpha|$, done. Suppose $\gamma \neq 0$. We show $\max(q|\beta|, r|\gamma|) \geq p|\alpha| = 2|\alpha|$; this forces the norm to equal $q|\beta|$ or $r|\gamma|$, hence a multiple of q or r .

Assume for contradiction that $2|\alpha| > r|\gamma|$ and $2|\alpha| > q|\beta|$. By the triangle inequality $|\beta| \geq |\alpha| - |\gamma|$, so $q|\beta| \geq q(|\alpha| - |\gamma|)$. The second assumption gives $q(|\alpha| - |\gamma|) < 2|\alpha|$, i.e. $(q - 2)|\alpha| < q|\gamma|$. The first assumption gives $|\gamma| < 2|\alpha|/r = 2|\alpha|/(q + 2)$. Substituting:

$$(q - 2)|\alpha| < q \cdot \frac{2|\alpha|}{q+2} \implies (q - 2)(q + 2) < 2q \implies q^2 - 4 < 2q \implies q^2 - 2q - 4 < 0.$$

For $q \geq 5$ this contradicts $q^2 \geq 2q + 5$ (formalized as `pasten_F32b_ineq` in the Lean kernel).

Exception. The triple $(2, 3, 5)$ ($q = 3 < 2p = 4$) is the unique exception: $\varphi = (8, -5, 3)$ gives norm 16, not a multiple of 3 or 5. For all other twin-prime triples ($q \geq 5$), the spectrum is exactly $\{kq\} \cup \{kr\}$ (verified numerically for $q \leq 73$, `t54`). $\square$

Corollary 33 (F32c: no Frobenius number for the non-degenerate spectrum). *For a twin-prime triple $(2, q, r = q + 2)$ with $q \geq 5$, the set of positive integers that are not non-degenerate norms is infinite. In particular, the non-degenerate norm spectrum has no Frobenius number.*

Proof. By Theorem 32, the non-degenerate norms are exactly $\{kq : k \geq 1\} \cup \{kr : k \geq 1\}$. For example, $q + r = 2q + 2$ is not divisible by either q or $q + 2$ (for $q \geq 5$), and the same is true for $m(q + r)$ whenever $\gcd(m, qr) = 1$. Thus infinitely many positive integers are absent from the spectrum. The Sylvester–Frobenius number $qr - q - r$ applies to the full numerical semigroup generated by q and r , not to this union of two rays; see also Remark 22. $\square$

Remark 22 (Semigroup structure). *The achievable non-degenerate norms $\{kq\} \cup \{kr\}$ form a proper subset of the numerical semigroup $\langle q, r \rangle$. Elements of $\langle q, r \rangle \setminus (\{kq\} \cup \{kr\})$, such as $q + r = 2q + 2$, are achieved only by degenerate vectors: the vector $\varphi = (k, k, 2k)$ satisfies $S_b = k = S_a$, giving $W = 0$. Theorem 32 shows that non-degeneracy exactly eliminates these mixed-term combinations, leaving only the pure scalings of q and r .*

Remark 23 (Note on all type $(1, 1, 1)$ triples having $p = 2$). *For any type $(1, 1, 1)$ triple (p, q, r) with $p < q$, we have $r = p + q$. If $p \geq 3$, both p and q are odd, so $r = p + q$ is even and composite. Therefore every type $(1, 1, 1)$ triple satisfies $p = 2$; the twin-prime condition q and $q + 2$ both prime applies universally.*

F33: second minimum non-degenerate norm for $\omega = 4$ type $(1, 1, 2)$

Theorem 34 (F33: second minimum for type $(1, 1, 2)$). *For type $(1, 1, 2)$ with $a = p$, $b = q$, $c = r_1 r_2$ ($p < r_1 < r_2$ primes, $p + q = r_1 r_2$, minimum non-degenerate norm $\text{nd} = r_1$):*

$$\text{second minimum non-degenerate norm} = \min(r_2, q, 2r_1).$$

Proof. Let $M = \min(r_2, q, 2r_1)$. We exhibit a vector of norm M and show no non-degenerate norm lies in (r_1, M) .

Achievability. If $M = r_2$: $\varphi = (1, 0, 0, 1)$ satisfies the constraint ($S_a + S_b = 1 = S_c$), is non-degenerate ($S_b = 0 \neq 1 = S_a$), with norm $\max(p, 0, 0, r_2) = r_2$.

If $M = q$: $\varphi = (1, -1, 0, 0)$ is non-degenerate with norm q .

If $M = 2r_1$: $\varphi = (-2, 0, -2, 0)$ is non-degenerate with norm $2r_1$.

Gap lemma. Suppose ψ is non-degenerate with $r_1 < \|\psi\|_\infty < M$. Since $r_2 \geq M$, we have $r_2|\varphi_{r_2}| < r_2$, so $\varphi_{r_2} = 0$. Since $q \geq M$, we have $q|\varphi_q| < q$, so $\varphi_q = 0$. Non-degeneracy ($\varphi_q \neq \varphi_p$) then forces $\varphi_p \neq 0$. The constraint gives $\varphi_p = \varphi_{r_1}$, so the norm equals $r_1|\varphi_{r_1}| = r_1|\varphi_p|$. For the norm to lie strictly in (r_1, M) we need $r_1 < r_1 k < M$ for some integer $k = |\varphi_{r_1}| \geq 2$. But $k \geq 2$ gives $r_1 k \geq 2r_1 \geq M$ (since $M \leq 2r_1$ by definition). Contradiction.

The integer core ($M \leq kr_1$ for $k \geq 2$ and $M \leq 2r_1$) is formalized as `pasten_F33_gap` in the Lean kernel (zero `sorry`). $\square$

Theorem 35 (F36: Universal second minimum for squarefree triples). *Let $a + b = c$ be a squarefree coprime triple with $\omega^* \geq 3$ distinct prime divisors $p_1 < p_2 < p_3 \leq \dots \leq p_{\omega^*}$. Then $\text{nd}(a, b) = p_2$ (Theorem 14) and*

$$\lambda_2(a, b) = \min(2p_2, p_3).$$

In particular: for $\omega^ = 3$ one always has $p_3 = p_1 + p_2 < 2p_2$, giving $\lambda_2 = p_3$ (recovering Theorem 31); for $\omega^* \geq 4$, either $2p_2 \leq p_3$ (gap formula $\lambda_2 = 2p_2$) or $p_3 < 2p_2$ (third-prime formula $\lambda_2 = p_3$).*

Proof. Lower bound. Suppose φ is a non-degenerate vector with $p_2 < \|\varphi\| < \min(2p_2, p_3)$. Since $p \cdot |\varphi_p| \leq \|\varphi\| < p_3 \leq p$ for every prime $p \geq p_3$, we have $|\varphi_p| < 1$, hence $\varphi_p = 0$ for all $p \geq p_3$. The constraint $\sum_i \alpha_i \varphi_i = 0$ then reduces to the two-variable equation $\alpha_{p_1} \varphi_{p_1} + \alpha_{p_2} \varphi_{p_2} = 0$ with $\alpha_{p_i} \in \{+1, -1\}$ (squarefree). The primitive solution is $\varphi_{p_1} = \pm 1$, $\varphi_{p_2} = \mp \alpha_{p_1} \alpha_{p_2}$, achieving norm $\max(p_1, p_2) = p_2$ (non-degeneracy holds since p_1 and p_2 lie in different prime groups when $\text{nd} = p_2$). Every solution is an integer multiple of this primitive vector; hence every achievable norm on this branch is kp_2 for $k \in \mathbb{Z}_{>0}$. No integer k satisfies $p_2 < kp_2 < 2p_2$, so no norm lies strictly in $(p_2, 2p_2)$. Together with $\|\varphi\| < p_3$, there is no norm in $(p_2, \min(2p_2, p_3))$.

Upper bound. Achieving $2p_2$: the nd-minimiser φ^* (with $\|\varphi^*\| = p_2$ and $W(\varphi^*) \neq 0$) satisfies $W(2\varphi^*) = 2W(\varphi^*) \neq 0$ and $\|2\varphi^*\| = 2p_2$, so $2p_2$ is an achievable non-degenerate norm.

Achieving p_3 : by the squarefree zero-coordinate property (Theorem 30), the nd-minimising sub-problem for the branch $\varphi_{p_1} = 0$ is a squarefree $(\omega^* - 1)$ -prime sub-triple whose primes are $\{p_2, \dots, p_{\omega^*}\}$. By Theorem 14 applied to this sub-triple, its minimum non-degenerate

norm equals p_3 (the second-smallest element of $\{p_2, \dots, p_{\omega^*}\}$), achieved by a vector φ' with $\varphi'_{p_1} = 0$, $\|\varphi'\| = p_3$, $W(\varphi') \neq 0$. Hence p_3 is achievable.

Since both $2p_2$ and p_3 exceed $\text{nd}(a, b) = p_2$, we get $\lambda_2 \leq \min(2p_2, p_3)$, completing the proof. $\square$

Remark 24. *Theorem 35 unifies F32a (Theorem 31), and Theorem 34 follows as a corollary: for type $(1, 1, 2)$ with $p < r_1 < r_2$ and $q = r_1 r_2 - p \gg r_2$, the sorted primes are $p_1 = p, p_2 = r_1, p_3 = r_2, p_4 = q$ and Theorem 35 gives $\lambda_2 = \min(2r_1, r_2)$; the “ q ” branch in Theorem 34 is always dominated by $r_2 \leq q$. T90b verifies Theorem 35 on 778 squarefree $\omega^* = 4$ triples with $a, b \leq 400$ (zero failures), plus 42 triples at $\omega^* = 5$ and 12 at $\omega^* = 6$ (all zero failures).*

Theorem 36 (F37: Third successive minimum for squarefree type $(1, 1, 2)$ triples). *Let $a + b = c$ be a squarefree coprime triple of type $(1, 1, 2)$ (i.e. $|P_a| = |P_b| = 1$, $|P_c| = 2$) with $\omega^* = 4$ distinct prime divisors $p_1 < p_2 < p_3 < p_4$. Then $\lambda_1 = p_2$ (E_n theorem), $\lambda_2 = \min(2p_2, p_3)$ (Theorem 35), and*

$$\lambda_3(a, b) = \begin{cases} \min(2p_2, p_4) & \text{if } p_3 < 2p_2 \quad (\text{Case 1: } \lambda_2 = p_3), \\ \min(3p_2, p_3) & \text{if } p_3 \geq 2p_2 \quad (\text{Case 2: } \lambda_2 = 2p_2). \end{cases}$$

Proof. Write $\lambda := \lambda_3(a, b)$. We prove matching lower and upper bounds in each case.

Upper bounds. In both cases $3p_2$ is achieved by tripling the nd -minimiser φ^* (non-degeneracy scales linearly). In Case 2, p_3 is achieved by the sub-problem with $\varphi_{p_1} = 0$: the E_n theorem applied to $\{p_2, p_3, p_4\}$ gives $\text{nd} = p_3$. In Case 1, $2p_2$ is achieved by doubling φ^* . The bound p_4 is achieved by an explicit non-degenerate vector with $|\varphi_{p_4}| = 1$: by the UD constraint $\varphi_{p_a} + \varphi_{p_b} = \varphi_{p_{c,1}} + \varphi_{p_{c,2}}$ (with $|P_a| = |P_b| = 1$, $|P_c| = 2$), setting the coordinate in the same UD-pair as p_4 to $+1$ and all other coordinates to 0 satisfies UD and gives $\varphi_{p_a} \neq \varphi_{p_b}$; the norm equals p_4 . Hence $\lambda \leq \min$ of the stated formula.

Lower bounds. Case 1 ($p_3 < 2p_2$, $\lambda_2 = p_3$). We show no non-degenerate norm lies in $(p_3, \min(2p_2, p_4))$. Let N be such a norm with $p_3 < N < \min(2p_2, p_4)$. Since $p_4 \cdot |\varphi_{p_4}| \leq N < p_4$, we get $\varphi_{p_4} = 0$. Since $p_3 \cdot |\varphi_{p_3}| \leq N < 2p_2$ and $|\varphi_{p_3}| \geq 2$ would give $p_3 \cdot |\varphi_{p_3}| \geq 2p_3 > 2p_2 > N$, we get $|\varphi_{p_3}| \leq 1$. If $|\varphi_{p_3}| = 1$: any coordinate with $p_2 |\varphi_{p_2}| > p_3$ requires $|\varphi_{p_2}| \geq 2$, giving $p_2 |\varphi_{p_2}| \geq 2p_2 > N$ — contradiction. So $|\varphi_{p_2}| \leq 1$ in either sub-case ($\varphi_{p_3} = 0$ or $|\varphi_{p_3}| = 1$). The UD constraint with $\varphi_{p_4} = 0$, $|\varphi_{p_3}| \leq 1$, $|\varphi_{p_2}| \leq 1$ forces $|\varphi_{p_1}| \leq 2$ (each group contributes at most 1 in absolute value to the UD sum), so $p_1 |\varphi_{p_1}| \leq 2p_1 < p_3 < N$ (three distinct primes satisfy $2p_1 < p_3$); thus p_1 does not determine the norm. The norm equals $p_2 |\varphi_{p_2}|$. For $N \in (p_3, 2p_2)$ this requires $p_3/p_2 < |\varphi_{p_2}| < 2$; since $p_3 > p_2$ gives $p_3/p_2 > 1$, no integer value of $|\varphi_{p_2}|$ exists. Contradiction; so N does not exist and $\lambda \geq \min(2p_2, p_4)$.

Case 2 ($p_3 \geq 2p_2$, $\lambda_2 = 2p_2$). We show no norm lies in $(2p_2, \min(3p_2, p_3))$. Let N be such a norm with $2p_2 < N < \min(3p_2, p_3)$. Since $N < p_3 \leq p_i$ for all $i \geq 3$, we have $\varphi_{p_3} = \varphi_{p_4} = 0$. The norm reduces to the rank-2 sub-lattice $\{p_1, p_2\}$ with achievable non-degenerate norms $\{kp_2\}$; no integer k satisfies $2p_2 < kp_2 < 3p_2$. Contradiction; so $\lambda \geq \min(3p_2, p_3)$. $\square$

Remark 25. *T94b verifies Theorem 36 on all 360 squarefree type- $(1, 1, 2)$ coprime triples with $c \leq 500$ (zero failures): 125 in Case 1 (including 71 where $p_4 < 2p_2$ so the p_4 -branch determines λ_3) and 235 in Case 2. The two cases are exhaustive and mutually exclusive; Case 1 reduces to $\lambda_3 = \min(2p_2, p_4)$ and Case 2 to $\lambda_3 = \min(3p_2, p_3)$, showing that λ_3*

is always determined by the “losing” branch of the λ_2 competition together with the next available prime.

F11: a conditional unboundedness criterion at $\omega \geq 6$

Theorem 37 (F11: conditional unboundedness criterion). *Let (n_a, n_b, n_c) be a partition type with $n_a + n_b + n_c = \omega \geq 6$. Suppose there is an infinite sequence of squarefree coprime triples of this type in which one constituent has at least three bounded prime factors and every prime outside that constituent tends to infinity while remaining within a fixed constant factor of the largest outside prime. Then*

$$\rho = \frac{\|\psi_{\text{nd}}\|_{\infty}}{R^{1/(\omega-1)}} \rightarrow \infty.$$

In particular, the corresponding partition type is unbounded along that sequence.

Proof. By Theorem 11, $\|\psi_{\text{nd}}\|_{\infty}$ is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$. Under the stated hypothesis, every finite group minimum other than the bounded constituent’s minimum tends to infinity; therefore the second-smallest finite group minimum is comparable to the outside scale N , and $\|\psi_{\text{nd}}\|_{\infty} \asymp N$. Since at least three primes remain bounded, $R = O(N^{\omega-3})$. Hence

$$\rho \gg \frac{N}{N^{(\omega-3)/(\omega-1)}} \rightarrow \infty. \quad \square$$

Remark 26. *For every $\omega \geq 6$ partition type, the required prime-pattern family is natural combinatorially and was checked on finite samples (for example, **t23** at $\omega = 6$), but its infinitude is not proved here. Thus the criterion is conditional and must not be read as an unconditional classification of every $\omega \geq 6$ type.*

F12–F16: upper bounds for bounded types

Combining the F10 formula with analytic optimization yields the following unconditional upper bounds for all universally-bounded types. Extremal constants that require an infinite prime-pattern family are conditional.

Theorem 38 (Upper bounds for bounded types). *Let $\rho = \|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$. For every squarefree coprime triple of each universally-bounded type, ρ is bounded as follows:*

ω	Type	upper bound	equality in bound?	source
3	(1, 1, 1)	$2^{-1/2} \approx 0.707$	No	F16
4	(0, 2, 2)	1	No	F14
4	(1, 1, 2)	$2^{-1/3} \approx 0.794$	No	F14
4	(1, 2, 1)	$2^{-1/3} \approx 0.794$	No	F12
5	(0, 2, 3)	1	No	F23
5	(0, 3, 2)	1	No	F23
5	(1, 2, 2)	$2^{-1/4} \approx 0.841$	No	F26
5	(2, 1, 2)	$2^{-1/4} \approx 0.841$	No	F30
5	(2, 2, 1)	$6^{-1/4} \approx 0.639$	No	F24

In particular, every bounded type with $a \geq 2$ has $\rho^{\omega-1} < 1/2$.

Proof. The $(0, 2, 3)$, $(0, 3, 2)$, $(1, 2, 2)$, and $(2, 2, 1)$ rows are Theorems 10, 8, and 9; the $(2, 1, 2)$ row follows from the $a \leftrightarrow b$ symmetry used in F30. The type- $(0, 2, 2)$ bound follows by applying the F10 formula to its two non-empty group minima. For the balanced rows, let $g \geq 2$ and d be the two smallest group minima. Every one of the $\omega - 2$ remaining primes is strictly larger than d , so $R > 2d^{\omega-1}$ and $\rho^{\omega-1} < 1/2$. For completeness, we spell out the basic type- $(1, 1, 1)$ estimate. For a prime triple (p, q, r) with $p \leq q \leq r$ and $p + q = r$, parity forces $p = 2$. Then $\rho^2 = q/(p(p + q)) = q/(2r)$. Since $q < r$: $\rho^2 < 1/2$. This is Lean-formalized (`pasten_F16_111_ratio_sq_lt_half`). Conditional sharpness would follow from infinitely many twin pairs $(q, q + 2)$; their infinitude is open. $\square$

Proof sketch of the type- $(0, 2, 2)$ row. If the two-prime constituents are pq and rs with one even prime, then nd is the minimum of the odd constituent. The other three prime factors include a prime larger than nd in its own constituent and the two primes of the even constituent, whose product exceeds $2nd$; hence $R > nd^3$ and $\rho < 1$. $\square$

Unifying pattern

The balanced types follow a single upper-bound formula.

Theorem 39 (Balanced-type universal upper bound, F16/F28). *For any partition type $(1, k_1, k_2)$ with $k_1 + k_2 = \omega - 1$ and any $\omega \geq 3$:*

$$\rho^{\omega-1} < \frac{1}{2},$$

equivalently $\rho < 2^{-1/(\omega-1)}$. The bound is never attained.

Proof. Let g be the smallest of the three group minima and let $d = nd$ be the second-smallest group minimum. Then $g \geq 2$, and the $\omega - 2$ primes other than g and d are all strictly greater than d : d is minimal in its group, while the third group minimum is at least d and distinct primes are unequal. Hence

$$\rho^{\omega-1} = \frac{d^{\omega-1}}{R} < \frac{d^{\omega-1}}{2 \cdot d \cdot d^{\omega-2}} = \frac{1}{2}. \quad \square$$

The constant would be sharp along a hypothetical infinite family in which the single-prime constituent is 2 and all other primes tend to equality with it. This is a prime-pattern condition, not an unconditional sharpness theorem. The inequalities were numerically checked for $\omega = 3, \dots, 7$.

General pattern

The classification for $\omega \in \{3, 4, 5\}$ exhibits a unified rule: the non-degenerate ratio $\|\psi_{nd}\|_{\infty}/R^{1/(\omega-1)}$ is *bounded* if and only if there exist two primes from opposite sign-sides (from $P_a \cup P_b$ vs. P_c , or from P_a vs. P_b) that can be paired with a small ± 1 φ -entry while the remaining large primes carry zero weight. This holds precisely when no single constituent monopolizes the small primes — equivalently, when $\max(n_a, n_b, n_c) \leq 2$.

6 Formal verification status

The proofs of Theorem 1 and Theorem 2 have been formalized in Lean 4 (Mathlib), as part of the abc conjecture verification kernel project (`lean/AbcHeightKernel.lean`, sections E3–E4, F3). Key items:

- `finite_prime_recip_sq_lt_one`: for any finite set P of primes, $\sum_{p \in P} p^{-2} < 1$. Proved by the telescoping argument above (zero `sorry`).
- `pasten_coeff_sq_sum`: the coefficient identity $\sum_p (R/p)^2 = R^2 \sum_p p^{-2}$.
- `pasten_det_lt_rad`: $\sqrt{\sum_p (R/p)^2} < R$.
- `minkowski_vaalder_pasten`: admitted as an axiom citing Vaaler [2]. Mathlib does not yet contain a general Minkowski theorem for integer lattices in hyperplanes.
- `pasten_prime_triple_arith` (F3): $p \cdot q < 4r$ for prime triple $p + q = r$, $p = 2$. Proved by `omega` after substitution.
- `pasten_rad_sqrt_lt_twice_r` (F3): $\sqrt{pqr} < 2r$ for prime triples. Proved from the above via `nlinarith` and `Real.sqrt_lt_sqrt`. Zero `sorry`.

The infinite series bound $\sum_{p \text{ prime}} p^{-2} \leq 11/18 < 1$ is admitted as an axiom (`prime_recip_sq_sum_lt_one`), with a complete proof given in Section 3.2 of the accompanying outsource document 0B-09.

The non-degeneracy results (F3–F10) are currently in the discovery tier (`discovery/m2_directions/`); F5 (lattice membership) and F8 (universal divisibility) have been formalized in Lean 4 (`lean/AbcHeightKernel.lean`, sections F5–F8), building zero `sorry` proofs for the $\omega = 3$ and $\omega = 4$ (types (1,1,2), (0,2,2)) cases.

The upper-bound core for type (1, 1, 1) (Theorem 39) has been formalized with two Lean 4 theorems (zero `sorry`):

- `pasten_F16_111_key`: $2q < p(p + q)$ for $p \geq 2$, $q > 0$ (proved by `nlinarith`).
- `pasten_F16_111_ratio_sq_lt_half`: $q/(p(p + q)) < 1/2$ in $\mathbb{R}$.

The upper-bound core for type (1, 2, 1) (F12) has been formalized:

- `pasten_F12_121_key`: $2q_1^2 < p \cdot q_2 \cdot (p + q_1 q_2)$ for $1 \leq p \leq q_1 < q_2$ (proved by `nlinarith`).
- `pasten_F12_121_ratio_cube_lt_half`: $q_1^2/(pq_2(p + q_1 q_2)) < 1/2$ in $\mathbb{R}$.

The upper-bound core for the c-even subfamily of type (1, 1, 2) (F14) has been formalized:

- `pasten_F14_112_ceven_key`: $2p^2 < q(p + q)$ for $1 \leq p < q$ (proved by `nlinarith`).
- `pasten_F14_112_ceven_ratio_cube_lt_half`: $p^2/(q(p + q)) < 1/2$ in $\mathbb{R}$.

These three families cover the $\omega \leq 4$ balanced-type upper-bound core.

For $\omega = 5$, the type $(1, 2, 2)$ key inequality is also formalized:

- `pasten_F27_122_rho4_key`: for $p+1 \leq q, r$ and $r+1 \leq s$, $p^3 < qrs$ (Nat, nlinarith).
- `pasten_F27_122_rho4_lt_half_real`: $p^3/(2qrs) < 1/2$ in $\mathbb{R}$ (Theorem 8 key inequality, Lean zero sorry).

For the universal pattern (Theorem 39), the general key lemma and the $\omega = 6$ instance are formalized:

- `pasten_F28_key_gen`: for $n \geq 1$ naturals each $> p$, $p^n < \prod x_i$ (Lean, `Finset.prod_le_prod + Nat.pow_lt_pow_left`).
- `pasten_F28_omega6_rho_lt_half`: $p^4/(2qrst) < 1/2$ in $\mathbb{R}$ ($\omega = 6$ instance of the universal pattern, zero sorry).

The general bound $\rho^4 < 1/6$ for all type $(2, 2, 1)$ triples (F29) is formalized:

- `pasten_F29_221_key`: $6q_1^3 < p_1p_2q_2r$ for $p_1 \geq 2$, $p_2 \geq p_1+1$, $q_2 \geq q_1+1$, $r \geq q_1q_2$ (Nat, `calc+nlinarith`).
- `pasten_F29_221_rho4_lt_sixth`: $q_1^3/(p_1p_2q_2r) < 1/6$ in $\mathbb{R}$ (`div_lt_one` pattern, zero sorry).

All proofs are zero sorry.

Additional lemmas added in the E10/E11/ E_n /F31 extension:

- `pasten_E10_lb_key`: if $\psi_q = \psi_r = 0$ and the $\omega = 3$ lattice constraint holds, then $W = 0$ (zero sorry).
- `pasten_E11_lb_key`: same for $\omega = 4$ type $(1, 1, 2)$ (zero sorry).
- `pasten_F31a`: $(a+2)^a < (2a)^{a+4}$ for all $a \geq 1$; algebraic core of the Quality- ρ Joint Bound (zero sorry).
- `pasten_quality_lt_half_core`: $a+b < ab$ for $a \geq 2$, $b \geq 3$; integer core of quality $< 1/2$ (zero sorry).
- `pasten_F32a_gap`: $q+2 \leq k \cdot q$ for all $k \geq 2$, $q \geq 3$; no non-degenerate norm lies strictly in $(q, q+2)$ (F32a core, zero sorry).
- `pasten_F32a_upper`: $\max(\max(0, q), q+2) = q+2$; norm of the second-minimum vector $\varphi = (0, 1, 1)$ (zero sorry).
- `pasten_F32b_ineq`: $2q+4 \leq q^2$ for $q \geq 5$; algebraic core blocking p -term dominance in the F32b spectrum proof (zero sorry).
- `pasten_F33_gap`: $N \leq k \cdot r_1$ for $k \geq 2$, $N \leq 2r_1$; integer core of second-minimum gap for type $(1, 1, 2)$ (F33, Theorem 34, zero sorry).

- `pasten_En_lb_general`: for any Finsets P_a, P_b, P_c of primes satisfying the lattice constraint $\sum_{P_a} f + \sum_{P_b} f = \sum_{P_c} f$, if $f = 0$ on P_b and $f = 0$ on P_c , then $\sum_{P_b} f = \sum_{P_a} f$ (degenerate condition for all ω ; Lean, zero `sorry`). This is the first fully general- ω Lean formalization in this project.
- Four OB-13B integer kernel lemmas (zero `sorry`): cross-group norm bound; $\text{median}^{\omega-1} \leq R$ for $\omega = 3, 4$; within-group $W \neq 0$; all in `lean/AbcHeightKernel.lean`.
- `pasten_nd_lb_large_prime_zero`: $p_2 \leq p, px < p_2 \Rightarrow x = 0$; integer core of LB-1 (zero `sorry`).
- `pasten_nd_lb_single_prime_int`: $k > 0, kz = 0 \Rightarrow z = 0$; abstract form of Lemma 18 (zero `sorry`).
- `pasten_nd_lb_omega2_ge_q`: $q \leq q(v/g)$ for $g \mid v$; integer core of $\text{nd} \geq p_2$ in Theorem 20 (zero `sorry`).
- Three type $(k, 1, 1)$ integer kernel lemmas for Theorem 23 (zero `sorry`):
`pasten_nd_k11_r_branch` ($r \leq rx$ for $x \geq 1$; $\varphi_r \neq 0$ branch);
`pasten_nd_k11_valuation_branch` ($qk \leq qkx$; $\varphi_r = 0$ branch); `pasten_nd_k11_norm_equals_qk`
 $(\max(px, qkx) = qkx \text{ for } p < q; q\text{-component dominates})$.
- `pasten_nd_km1_gcd_div_one`: $\gcd(k/g, m/g) = 1$ for $g = \gcd(k, m)$; integer core of the $\varphi_r = 0$ solution-family uniqueness in Theorem 25 (Mathlib `Nat.Coprime.div_gcd_coprime`, zero `sorry`).
- `pasten_nd_kmn_val_witness_constraint`: $k \cdot (m/g) = m \cdot (k/g)$ for $g = \gcd(k, m)$; confirms the valuation-regime witness $(-m/g, k/g, 0)$ satisfies the constraint for *any* n (Corollary 26), using `Nat.mul_div_assoc` twice. Zero `sorry`.
- `pasten_nd_kmn_val_n_zero`: $n \cdot 0 = 0$ for all n ; captures the key n -independence: the r -coordinate of the valuation-regime witness is zero, so the c -exponent n does not appear (Corollary 26). Zero `sorry`.

7 Conclusion

We have proved that for squarefree coprime triples (a, b, c) with $a + b = c$ and $\omega(abc) \geq 2$, the Pasten lattice satisfies $\det(F(a, b)) < \text{rad}(abc)$, and hence Vaaler's subspace lattice theorem [2] gives a nonzero vector with $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$.

Beyond the determinant bound, we have established a structural classification of non-degeneracy for all partition types with $\omega \leq 5$, proved that all types with $\omega \geq 6$ satisfy the criterion of Theorem 37 whenever the required prime-pattern family exists, and proved unconditional upper bounds for every universally-bounded type (Theorems 38 and 39). The central structural results are: (a) the Universal Divisibility Lemma ($q \mid \psi_q$ for all $q \in P$, proved by elementary divisibility from the lattice constraint); (b) the φ -coordinate change reducing $F(a, b)$ to the simplest integer hyperplane lattice; (c) the optimal crossing formula (Theorem

F10): the minimum non-degenerate norm is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$, achieved by any cross-group prime pair; (d) the universal lower bound (Theorem E_n): every non-degenerate ψ satisfies $\|\psi\|_\infty \geq \text{nd}$, proved by a clean F8+constraint+Wronskian argument and Lean-formalized for $\omega = 3, 4$ (E10/E11), numerically verified for all types $\omega \leq 7$; (e) a complete table of non-degeneracy upper bounds for all partition types $\omega \in \{3, 4, 5\}$ (Table in Theorem 38); (f) the unifying upper bound $\rho^{\omega-1} < 1/2$ for *all* balanced types and $\omega \geq 3$ (Theorem 39, F28), with the general key lemma Lean-formalized; and (g) the general bound $\rho^4 < 1/6$ for *all* type $(2, 2, 1)$ triples (F29, Lean-formalized), with conditional sharpness at $6^{-1/4}$; and (h) the corrected bound $\rho^4 < 1/2$ for *all* type $(2, 1, 2)$ triples (F30, Lean-formalized, correcting F25), with conditional sharpness equal to the $(1, 2, 2)$ constant by $a \leftrightarrow b$ symmetry; and (i) the Quality- ρ Joint Bound (F31, Theorem 5): for every non-trivial squarefree triple of any bounded- ρ type, $\text{quality} + \rho^{\omega-1} < 1$, proved by combining $\text{quality} < 1/2$ (Proposition 4) with $\rho^{\omega-1} < 1/2$ (Theorems F12–F30); and (j) the second-minimum and complete spectrum results for type $(1, 1, 1)$: the second minimum non-degenerate norm equals r (F32a, Theorem 31), and for every twin-prime triple with $q \geq 5$ the full spectrum of achievable non-degenerate norms is exactly $\{kq : k \geq 1\} \cup \{kr : k \geq 1\}$ (F32b, Theorem 32), both Lean-formalized; this union of two rays has no Frobenius number (Corollary 33); and second minimum $\min(r_2, q, 2r_1)$ for type $(1, 1, 2)$ (F33, Theorem 34), Lean-formalized; and (k) the non-squarefree upper bound (OB-13B, Theorem 15): $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$ for all coprime triples, proved via the cross-group vector construction, with a tighter median form (Corollary 16), and the within-group construction showing that the squarefree E_n lower bound fails for non-squarefree triples (counterexample (72, 11, 83)); (l) the universal lower bound (Theorem 19): $\text{nd}(a, b) \geq p_2$ (second-smallest prime in $\text{rad}(abc)$) for all coprime triples, squarefree or not, proved by a single-prime-constraint argument (Lemma 18), tight at $(1, 2, 3)$; and (m) the exact formula for $\omega^* = 2$ triples (Theorem 20, Corollary 21): $\text{nd}(a, b) = \max(p \cdot w/g, q \cdot v/g)$ where $g = \gcd(v, w)$, verified on all seven $\omega^* = 2$ examples up to $c \leq 1000$, Lean-formalized; and (ag) the complete norm spectrum for $\omega^* = 2$ triples (Corollary 22): the achievable norms of non-degenerate vectors are exactly $\{k \cdot \text{nd}(a, b) : k \in \mathbb{Z}_{>0}\}$, so all successive minima satisfy $\lambda_k = k \cdot \text{nd}$ —the lattice is effectively one-dimensional and its spectrum is a pure arithmetic progression; and (n) the exact formula for type $(k, 1, 1)$ non-squarefree triples (Theorem 23): $\text{nd}(a, b) = \min(r, qk)$ for $P_a = \{p^k\}$, $P_b = \{q\}$, $P_c = \{r\}$ with $p < q < r$, with two regimes (valuation-dominant $qk < r$ and pairwise-dominant $qk \geq r$) and confirming $\text{nd} \geq p_2 = q$ (Theorem 19); and (ah) the second successive minimum for type $(k, 1, 1)$ triples (Theorem 24): in the valuation regime $\lambda_2 = \min(r, 2qk)$; in the pairwise regime $\lambda_2 = \min\{N \in \{2r, qk, q(k-1)\} : N > r\}$ (three explicit achievability vectors, gap proved by a parametric line argument); verified on 55 triples, zero failures (T87); verified numerically (T68) and Lean-formalized (three arithmetic kernel lemmas, zero `sorry`); and (ai) the second successive minimum for type $(k, m, 1)$ triples in the valuation regime (Theorem 27): $\lambda_2 = \min(r, 2N_0)$ where $N_0 = \max(pm/g, qk/g)$, $g = \gcd(k, m)$, proved by the same three-way split as Theorem 24 (the $\varphi_r = 0$ branch generates norms $\{tN_0 : t \geq 1\}$ by primitivity of its generator, any $|\varphi_r| \geq 1$ vector gives norm $\geq r \geq \min(r, 2N_0)$); verified on 40 triples ($p \leq 7$, $k, m \in \{2, \dots, 9\}$, zero failures, T88b); and (o) the exact formula for general type $(k, m, 1)$ non-squarefree triples (Theorem 25): $\text{nd}(a, b) = \min(r, \max(pm/g, qk/g))$ where $g = \gcd(k, m)$, subsumes Theorems 23 ($m = 1$) and the squarefree E_n value ($k = m = 1$), with equal-valuation regime ($k = m$) always

realising $\text{nd} = p_2$; verified numerically on 14 named examples (T69); and (p) a universality corollary (Corollary 29): for one-prime-per- a, b -group triples with $r_1 = \min P_c > q > p$, the formula $\text{nd} = \min(r_1, \max(pm/g, qk/g))$ holds regardless of how many primes c has or their valuations — only the smallest c -prime matters; and (q) a valuation-regime independence result (Corollary 26): when $\max(pm/g, qk/g) \leq r$, the formula $\text{nd}(a, b) = \max(pm/g, qk/g)$ holds for *all* $n = v_r(c)$, not just $n = 1$ — the witness $(-m/g, k/g, 0)$ satisfies the constraint for any n , so the c -exponent is entirely invisible in the valuation regime (verified numerically on 52 cases with $n \geq 2$, T71); and (r) a characterisation of the pure-formula boundary (Remark 10): when the group- c prime is not the largest in the triple, the three-pure formula can fail and a fully mixed witness is required, exemplified by $(5, 27, 32)$ where $\text{nd} = 6$ beats all pure-branch minima ($N_0 = N_1 = 15$, $N_2 = 25$); and (s) a complete pairwise-regime characterisation (Theorem 28): for one-prime-per-group triples in the pairwise regime $N_{p_L=0} > p_L$, the exact formula is $\text{nd}(a, b) = \min(N, \max(p_L, B))$ where B is the minimum weighted norm over $|\varphi_{p_L}| = 1$ solutions — simultaneously giving $\text{nd} = p_L$ in the typical case ($B \leq p_L$, verified 223/229 cases, T72) and recovering the precise value in the 6 exceptional cases (e.g. $(169, 343)$ with $\text{nd} = N = 21$ due to a gcd obstruction); and (t) a systematic scan (Remark 12, T73) of all quality- > 1 one-prime-per-group triples with $a + b \leq 5 \times 10^5$, yielding exactly 10 such triples — all correctly classified by the exact formula, with four exhibiting $\text{nd} < p_L$ via the $\text{val}(\varphi_{p_L} = 0)$ regime, and confirming that nd and quality are structurally independent (nd depends on (p_a, p_b, p_c, k, m) while quality depends on n); Items (u)–(ae) below concern the valuation-coordinate lattice F_{val} and are candidate formulas with finite all-nonzero-case verification; they are not unconditional theorems and do not determine the original Pasten lattice minimum in general. and (u) the exact nd formula for $\omega^* = 4$ triples with two c -group primes (Candidate Formula 1): for $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r, s\}$, $\text{nd}(a, b) = \min(\text{nd}_3(p^k, q^m, r^j), \text{nd}_3(p^k, q^m, s^n), \max(q, B_m), \max(p, B_k))$ where B_x is the two-dimensional Bezout minimum for $j\varphi_r + n\varphi_s = x$; the $\varphi_q = 0$ branch wins in 25 out of 431 tested cases (T74), confirming that the four-branch formula subsumes the $\omega^* = 3$ characterization; and (v) the exact nd formula for type $(2, 1, 1)$ triples (Candidate Formula 2): for $P_a = \{p, q\}$, $P_b = \{r\}$, $P_c = \{s\}$, the minimum is taken over four branches corresponding to zeroing one prime-group slot — $N_{s=0}$ ($\text{Pa} \times \text{Pb}$ interaction, subsumes within-group and pairwise norms), $N_{r=0}$ ($\text{Pa} \times \text{Pc}$ interaction), and two $\omega^* = 3$ sub-problems $\text{nd}_3(p^{k_1}, r^m, s^n)$, $\text{nd}_3(q^{k_2}, r^m, s^n)$; verified on 672 triples (T77, 0 failures), with $N_{s=0}$ dominant in 87% of cases; and (w) the exact nd formula for type $(1, 2, 1)$ triples (Candidate Formula 3): for $P_a = \{p\}$, $P_b = \{q, r\}$, $P_c = \{s\}$, the same four-branch structure with $P_a \leftrightarrow P_b$ exchanged — $N_{s=0}$ ($\text{Pb} \times \text{Pa}$ interaction), $N_{p=0}$ ($\text{Pb} \times \text{Pc}$ interaction), and two $\omega^* = 3$ sub-problems; verified on 1 043 triples (T78_formula, 0 failures), with $N_{s=0}$ dominant in 90% of cases; and (x) the exact nd formula for type $(2, 2, 1)$ triples (Candidate Formula 4): for $P_a = \{p, q\}$, $P_b = \{r, s\}$, $P_c = \{u\}$, the minimum is taken over five branches — $N_{u=0}$ (the P_a – P_b homogeneous interaction, zeroing φ_u), two type- $(1, 2, 1)$ sub-problems (zeroing φ_p or φ_q via Candidate Formula 3), and two type- $(2, 1, 1)$ sub-problems (zeroing φ_r or φ_s via Candidate Formula 2); verified on 1 425 triples with $a, b \leq 300$ (T79, spot-check 60, 0 failures), with $N_{u=0}$ dominant in 98.6% of cases; and (y) the exact nd formula for type $(1, 2, 2)$ triples (Candidate Formula 5): for $P_a = \{p\}$, $P_b = \{q, r\}$, $P_c = \{s, t\}$, the minimum over five branches — $N_{p=0}$ (P_b – P_c homogeneous interaction, zeroing φ_p), two type- $(1, 1, 2)$ sub-problems (zeroing φ_q or φ_r via Candidate Formula 1), and two type- $(1, 2, 1)$ sub-problems (zeroing φ_s or φ_t

via Candidate Formula 3); verified on 996 triples with $a, b \leq 150$ (T80, spot-check 50, 0 failures), with $N_{p=0}$ dominant in 87.2% of cases; and (z) the exact nd formula for type (2, 1, 2) triples (Candidate Formula 6), obtained from Candidate Formula 5 by the $P_a \leftrightarrow P_b$ label swap: $N_{r=0}$ (P_a - P_c interaction) plus two type-(1, 1, 2) and two type-(2, 1, 1) sub-problems; and (aa) the exact nd formula for type (1, 1, 3) triples (Candidate Formula 7): for $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r, s, t\}$, the minimum of five branches — three type-(1, 1, 2) sub-problems (zeroing each c -prime in turn via Candidate Formula 1) and two one-sided norms $N_{p=0}$, $N_{q=0}$; verified on 729 triples with $a, b \leq 300$ (T81, spot-check 60, 0 failures), with the $\varphi_t = 0$ branch dominant (91.8%); and (ab) the exact nd formula for type (1, 3, 1) triples (Candidate Formula 8): for $P_a = \{p\}$, $P_b = \{q, r, s\}$, $P_c = \{t\}$, the minimum of five branches — a $P_a \times P_b$ homogeneous norm $N_{t=0}$ (zeroing φ_t , $W \neq 0$), a $P_b \times P_c$ interaction norm $N_{p=0}$ (zeroing φ_p , $\varphi_q + \varphi_r + \varphi_s \neq 0$), and three type-(1, 2, 1) sub-problems (zeroing each b -prime via Candidate Formula 3); verified on 843 triples with $a, b \leq 300$ (T82, spot-check 60, 0 failures), with $N_{t=0}$ dominant (95.3%); and (ac) the exact nd formula for type (3, 1, 1) triples (Candidate Formula 9), obtained from Candidate Formula 8 by the $P_a \leftrightarrow P_b$ label swap; the mirror case was also checked directly on all 843 type-(3, 1, 1) triples with $a, b \leq 300$ (T82-type311, 0 failures); and (ad) the *zero-coordinate minimizer property* (Candidate Formula 10): for $\omega^* \geq 4$, the minimum non-degenerate valuation-lattice norm is achieved by a vector with at least one prime coordinate equal to zero in all surveyed cases, so $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ on those samples (minimum over all zero- φ_p sub-lattices); the squarefree case is proved, while $\omega^* = 4, 5, 6$ non-squarefree cases are finite-verification claims: non-squarefree $\omega^* = 6$: 1959 triples $a, b \leq 200$, all three partition classes $\{4, 1, 1\}$, $\{3, 2, 1\}$, $\{2, 2, 2\}$, 0 violations (T85bis); squarefree $\omega^* = 6$: proved by Theorem 30. This fails for non-squarefree $\omega^* = 3$: the triple (5, 27, 32) achieves $\text{nd}_{\text{val}} = 6$ via an all-nonzero vector, while every zero-coordinate branch gives norm ≥ 15 ; and (ae) the *general recursive valuation-nd algorithm* (Candidate Formula 11): for any coprime triple with $\omega^* \geq 3$ primes, the candidate formula is $\text{nd}_{\text{val}}(a, b) = \min_p B_p$ (each B_p zeroing one prime coordinate, computed recursively from the one-prime-per-group base case); it is proved in the squarefree case, and verified on finite samples for non-squarefree $\omega^* = 4, 5, 6$, including $\omega^* = 5$ (all six partition types) and $\omega^* = 6$ (all three partition classes, non-squarefree 1959 triples $a, b \leq 200$, 0 violations, T85bis; squarefree case from Theorem 30); and (af) the *squarefree minimizer support theorem* (Theorem 30): for every squarefree coprime triple of any $\omega^* \geq 3$, every nd-minimizer satisfies $\varphi_p = 0$ for all $p > m_2$ (second group minimum), so at least $\omega^* - 2$ coordinates are forced to zero by integrality; in particular the zero-coordinate property holds unconditionally for all squarefree triples; and (aj) the universal second-minimum formula for all squarefree triples with $\omega^* \geq 3$ (Theorem 35): $\lambda_2(a, b) = \min(2p_2, p_3)$ where $p_1 < p_2 < p_3$ are the three smallest primes of $\text{rad}(abc)$, proved by an integrality gap argument reducing any near-minimizer to the two-prime sub-lattice; unifies Theorem 31 ($\omega^* = 3$, where $p_3 < 2p_2$ always) and Theorem 34 (type (1, 1, 2), where the q -branch is dominated); verified on 778 squarefree $\omega^* = 4$ triples, 42 at $\omega^* = 5$, 12 at $\omega^* = 6$ (zero failures, T90b); and (ak) the third successive minimum formula for squarefree type-(1, 1, 2) triples (Theorem 36): $\lambda_3(a, b) = \min(2p_2, p_4)$ if $p_3 < 2p_2$, and $\lambda_3(a, b) = \min(3p_2, p_3)$ if $p_3 \geq 2p_2$, where p_4 is the fourth-smallest prime of $\text{rad}(abc)$; proved by extending the integrality gap argument to the next branch — the “losing” branch of the λ_2 competition always yields λ_3 ; verified on 360 squarefree type-(1, 1, 2) triples with $c \leq 500$ (zero failures, T94b).

The non-degenerate ratio $\|\psi_{\text{nd}}\|/R^{1/(\omega-1)}$ is bounded for exactly the “balanced” partition types (both other constituents non-empty with available small-prime crossings), and grows without bound for the “isolated” types where both non-dominant constituents are forced-large single primes. For $\omega \geq 6$, unboundedness follows from Theorem 37 once the required prime-pattern family exists. The unconditional upper bounds for the bounded types are collected in Theorem 38. Their sharpness depends on corresponding infinite prime-pattern families and remains conditional.

We also prove a quality boundary for the squarefree subfamily (Proposition 4): quality $> 1/2$ if and only if $a = 1$. Within bounded partition types, quality and ρ are negatively correlated ($r \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$): the triples closest to the extremal ρ bound have quality $\rightarrow 1/3$, while the highest-quality non-trivial squarefree triples ($a = 2$, $b \rightarrow \infty$) have $\rho \rightarrow 0$. This quality- ρ trade-off is a provable structural feature, not a numerical artifact.

Honest scope. None of these results constitute progress toward the abc conjecture or SDC. The Minkowski bound is $\eta_R = 1/(\omega - 1)$ relative to $\text{rad}(abc)$, whereas SDC requires $\eta_c < 1$ relative to c for *all* $\varepsilon > 0$ — an exponential improvement. The classification of degenerate vs. non-degenerate types is structural geometry of Pasten’s lattice; it does not produce bounds on c . The abc conjecture remains open.

This is a structural result about the geometry of Pasten’s lattice construction, not progress toward abc. The abc conjecture requires $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ (Pasten’s SDC), and the gap between our bound and SDC is the full depth of the problem. We have made no use of any abc-equivalent hypothesis, known abc triples, or the Szpiro conjecture.

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